

Radiator Analysis and Sizing

Kevin Martinez, Keaton Huffman, Alexander Quinonez, Nicholas Rios, Ryan Rivell, Maanav Venkatasubramanian

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Abstract

This report analyzes the radiator cooling system used on the University of Texas at Arlington FSAE F26 racecar to determine whether the current dual-radiator configuration is necessary or if a single-radiator configuration could provide comparable cooling performance. The analysis begins with a thermodynamic energy balance of the turbocharged Yamaha R3 engine using E85 fuel to estimate the required coolant heat rejection rate. From the calculated fuel energy input, the cooling system was estimated to reject approximately 48.3 kW of heat under peak-load conditions, corresponding to about 24.1 kW per radiator for the current dual-radiator setup. The report then compares the current F26 radiator configuration to the previous F20R single-radiator configuration using core geometry, tube count, fin spacing, air-side heat transfer correlations, coolant-side internal flow analysis, and the effectiveness-NTU method. A simplified Ansys Fluent conjugate heat transfer model was also developed to study heat transfer through a representative section of the radiator and to support future validation of the analytical model. In addition, an Excel-based radiator sizing tool was created to allow radiator geometry, fluid properties, heat rejection requirements, and pressure limits to be compared during future design iterations. The results of this work provide a structured framework for evaluating radiator performance and identifying the key parameters needed to determine whether the F26 cooling system can be simplified while maintaining adequate thermal performance during endurance operation.

Nomenclature

c_p	Specific heat of exhaust gas ($\frac{Btu}{lbm \cdot ^\circ F}$)
T_{exh}	Exhaust temperature ($^\circ F$)
T_{ref}	Ambient reference temperature ($^\circ F$)
$\dot{Q}_{fuel}$	Fuel chemical energy rate (Btu)
$\dot{W}_b$	Engine brake power (Btu)
$\dot{Q}_{coolant}$	Heat transferred to engine coolant (Btu)
$\dot{Q}_{oil}$	Heat transferred to engine oil (Btu)
$\dot{Q}_{exhaust}$	Energy remaining in exhaust stream (Btu)
$\dot{Q}_{other}$	Minor radiation and mechanical losses (Btu)
p_{tube}	Tube pitch (in)
p_{fin}	Fin pitch (in)
σ	Airflow face area ratio (-)

A_{ht}	Air-side heat transfer area (in^2)
H_{air}	Air gap (in)
V_{eff}	Effective velocity
G	Mass flux ($\frac{lbm}{ft^2 \cdot s}$)
h_x	Convection coefficient ($\frac{Btu}{hr \cdot ft^2 \cdot ^\circ F}$)
η_{fin}	Fin efficiency (-)
m	Fin parameters (in^{-1})
η_o	Overall surface efficiency (-)
$D_{h,w}$	Wetted hydraulic diameter (in)
Re_x	Reynolds number (-)
Nu_x	Nusselt number (-)
U	Overall Heat transfer coefficient ($\frac{Btu}{hr \cdot ft^2 \cdot ^\circ F}$)
AR	Area ratio (-)
C_x	Heat capacity rate ($\frac{Btu}{hr \cdot ^\circ F}$)
ε_x	Effectiveness (-)

I. Introduction

The cooling system is a critical part of the UTA FSAE F26 racecar because it must remove enough heat from the engine to maintain stable operating temperatures during endurance conditions. The F26 car uses a turbocharged Yamaha R3 engine running on E85 fuel, which increases the importance of controlling coolant temperature due to the added thermal demands from combustion, exhaust energy, and turbocharger operation. If the radiator system cannot reject heat at a sufficient rate, coolant temperatures will rise, reducing engine reliability and overall vehicle performance.

The current F26 cooling system uses two dual-pass radiators packaged inside the sidepods. These sidepods help guide airflow toward the radiators while also protecting the cores from debris and reducing packaging conflicts with other vehicle systems. However, the use of two radiators adds weight, plumbing complexity, and potential pressure losses. Because of this, the main objective of this report is to determine whether the current dual-radiator configuration is necessary or if a single-radiator configuration could provide comparable cooling performance.

To evaluate this, the report uses a combination of thermodynamic energy balance, heat exchanger analysis, numerical modeling, and simplified Ansys Fluent simulation. First, the required heat rejection rate is estimated using the fuel energy input from E85 and the average endurance operating condition from MoTeC data. Then, the current F26 radiator configuration is compared to the previous F20R single-radiator configuration using radiator geometry, tube count, fin spacing, coolant-side heat transfer, air-side heat transfer, and the effectiveness-NTU method. This comparison helps determine how core area, airflow, coolant flow rate, and radiator geometry affect overall cooling performance.

A simplified Ansys Fluent conjugate heat transfer model is used to study heat transfer through a representative section of the radiator and support the trends predicted by the numerical analysis. In addition, an Excel-based

radiator sizing tool was developed to make future radiator comparisons easier by allowing the user to change thermal, fluid, geometric, and pressure inputs.

The goal of this report is not only to compare the current and previous radiator configurations, but also to establish a repeatable method for future cooling system design. By combining endurance data, analytical heat transfer methods, and simulation, this study provides a framework for determining whether a radiator configuration can meet the thermal demands of the F26 racecar while also supporting packaging, aerodynamic, and reliability goals.

II. Numerical Analysis

A numerical analysis was performed to estimate the heat rejection requirement for the cooling system on the F26 racecar. The purpose of this section is to establish a representative radiator design condition based on average endurance operation. Since the car does not operate at maximum power continuously during an endurance event, the final radiator sizing condition was based on the average endurance horsepower rather than the maximum engine horsepower. A peak-load calculation was still included as an upper-bound reference and was then scaled to the average endurance condition.

The analysis begins with the total chemical energy supplied by the E85 fuel. Since E85 is a blend of ethanol and gasoline, its lower heating value was estimated using the lower heating values and densities of ethanol and gasoline. E85 was assumed to be composed of 85% ethanol and 15% gasoline by volume. Because the lower heating values are given per unit mass, the volume fractions were first converted into masses.

From the fuel property table [1],

$$\begin{aligned} LHV_{ethanol} &= 11,530 \frac{Btu}{lbm} \\ LHV_{gasoline} &= 18,900 \frac{Btu}{lbm} \\ \rho_{ethanol} &= 49.3 \frac{lbm}{ft^3} \\ \rho_{gasoline} &\approx 47 \frac{lbm}{ft^3} \end{aligned}$$

Assuming a total E85 mixture volume of $1 ft^3$

$$\begin{aligned} V_{ethanol} &= 0.85 ft^3 \\ V_{gasoline} &= 0.15 ft^3 \end{aligned}$$

The mass of ethanol in the mixture is

$$\begin{aligned} m_{ethanol} &= (0.85)(49.3) \\ m_{ethanol} &= 41.91 lbm \end{aligned}$$

The mass of gasoline in the mixture is

$$\begin{aligned} m_{gasoline} &= (0.15)(47) \\ m_{gasoline} &= 7.05 lbm \end{aligned}$$

Therefore, the total mass of the E85 mixture is

$$\begin{aligned} m_{total} &= 41.91 + 7.05 \\ m_{total} &= 48.96 lbm \end{aligned}$$

The energy contribution from the ethanol is

$$\begin{aligned}Q_{ethanol} &= (41.91)(11,530) \\Q_{ethanol} &\approx 483,200 \text{ Btu}\end{aligned}$$

The energy contribution from the gasoline is

$$\begin{aligned}Q_{gasoline} &= (7.05)(18,900) \\Q_{gasoline} &\approx 133,200 \text{ Btu}\end{aligned}$$

The total energy contained in the mixture is

$$\begin{aligned}Q_{total} &= 483,200 + 133,200 \\Q_{total} &\approx 616,400 \text{ Btu}\end{aligned}$$

The lower heating value of E85 is then

$$\begin{aligned}LHV_{E85} &= \frac{Q_{total}}{m_{total}} \\LHV_{E85} &= \frac{616,400}{48.96} \\LHV_{E85} &\approx 12,600 \frac{\text{Btu}}{\text{lbm}}\end{aligned}$$

This value was used as the lower heating value for E85 in the numerical energy balance. The fuel energy input rate is calculated using the fuel mass flow rate and the lower heating value of E85:

$$\dot{Q}_{fuel} = \dot{m}_f LHV$$

Using an estimated fuel mass flow rate of 52 lbs/hr

$$\dot{m}_f = 0.0144 \frac{\text{lbm}}{\text{s}}$$

and the estimated E85 lower heating value of

$$LHV_{E85} = 12,600 \frac{\text{Btu}}{\text{lbm}}$$

The fuel energy input rate at the peak-load condition becomes

$$\begin{aligned}\dot{Q}_{fuel} &= (0.0144)(12,600) \\ \dot{Q}_{fuel} &\approx 181.9 \frac{\text{Btu}}{\text{s}}\end{aligned}$$

Converting to Btu/hr,

$$\begin{aligned}\dot{Q}_{fuel} &= 181.9(3600) \\ \dot{Q}_{fuel} &\approx 655,000 \frac{\text{Btu}}{\text{hr}}\end{aligned}$$

This value represents the approximate chemical energy entering the engine per hour at the peak-load condition. The maximum engine power output is estimated to be

$$P_{max} = 69 \text{ HP}$$

However, the racecar does not operate at maximum power continuously during endurance. From the endurance run data, the average power output was approximately 17 HP. After removing the low-power portion at the beginning of the data set, the representative average endurance power was assumed to be closer to

$$P_{avg} = 20 \text{ HP}$$

Therefore, the 69 HP condition was used as a peak reference case, while the 20 HP condition was used as the main radiator design condition.

The steady-state energy balance for the engine can be written as

$$\dot{Q}_{fuel} = \dot{W}_b + \dot{Q}_{coolant} + \dot{Q}_{exhaust} + \dot{Q}_{oil} + \dot{Q}_{other}$$

where $\dot{W}_b$ is the useful brake power, $\dot{Q}_{coolant}$ is the heat rejected to the coolant, $\dot{Q}_{exhaust}$ is the heat carried away by the exhaust, $\dot{Q}_{oil}$ is the heat absorbed by the engine oil, and $\dot{Q}_{other}$ includes smaller losses such as radiation, friction, and other mechanical losses.

For this analysis, the coolant heat rejection was estimated using a representative fraction of the fuel energy input. For a spark-ignition engine, approximately 20–30% of the fuel's chemical energy may be rejected to the coolant. To remain conservative, a value of 30% was used:

$$\dot{Q}_{coolant,max} = 0.30\dot{Q}_{fuel}$$

Substituting the peak fuel energy rate,

$$\begin{aligned}\dot{Q}_{coolant,max} &= 0.30(655,000) \\ \dot{Q}_{coolant,max} &\approx 196,500 \frac{\text{Btu}}{\text{hr}}\end{aligned}$$

This value represents the estimated coolant heat rejection at the peak-load condition. Since the radiator is being evaluated under average endurance conditions, the peak coolant heat rejection was scaled by the ratio of average endurance power to maximum engine power. This assumes that coolant heat rejection varies approximately linearly with engine output over the operating range of interest.

$$\dot{Q}_{coolant,avg} = \dot{Q}_{coolant,max} \left(\frac{P_{avg}}{P_{max}} \right)$$

Substituting

$$P_{avg} = 20 \text{ HP}$$

and

$$P_{max} = 69 \text{ HP}$$

gives

$$\dot{Q}_{coolant,avg} = 196,500 \left(\frac{20}{69} \right)$$

$$\dot{Q}_{coolant,avg} \approx 57,000 \frac{Btu}{hr}$$

Therefore, under the representative average endurance condition, the engine cooling system must reject approximately

$$\dot{Q}_{coolant,avg} \approx 57,000 \frac{Btu}{hr}$$

through the coolant system.

Since the F26 racecar uses two radiators, the total heat rejection load was assumed to be evenly distributed between the two units for this preliminary estimate. Therefore, the heat rejection requirement per radiator is

$$\dot{Q}_{rad} = \frac{\dot{Q}_{coolant,avg}}{2}$$

$$\dot{Q}_{rad} = \frac{57,000}{2}$$

$$\dot{Q}_{rad} \approx 28,500 \frac{Btu}{hr}$$

Thus, each radiator should be capable of rejecting approximately

$$\dot{Q}_{rad} \approx 29,000 \frac{Btu}{hr}$$

under the average endurance design condition. This value establishes the main thermal design target for the radiator analysis. The peak-load calculation was retained as an upper-bound reference, but the average endurance condition was selected as the primary design case because it better represents the sustained operating condition experienced during an endurance event.

III. Performance Analysis of F26 and F20R Radiator Selection

The current radiator configuration for the UTA Racing F26 FSAE car is going to use a mirrored dual-pass radiator setup in series. The core of these radiators are 6" in width and 9" in length. This results in a 54 in² core area per radiator, netting a total of 108 in². F20R is a single dual-pass radiator, approximately to be about the same frontal core area. A claim is that only one radiator is needed, or rather that the single radiator configuration is superior to the two-radiator design. The goal of this section is to either prove or disprove this claim through heat transfer analysis. The analysis of the automotive radiator becomes particularly complicated due to the intricacies of the fin geometry, variable mass flow rates, and temperature variation. Thus, a few assumptions must be made for a detailed analysis, especially without the availability of data. The radiator will also be analyzed using the following assumptions:

1. The radiator must be analyzed at the average (heat-soaked) condition.
2. The inlet temperature of the coolant is fixed at the average condition.
3. Zero to little ram air is flowing through the sidepods.
4. Evaluate the system using the mechanical pump only.
5. Analyze the performance of the core only.

The average conditions were collected using MoTeC data from a previous endurance race. This data is quite limited, as only the coolant temperature going into the radiator and the air temperature directly after the fan were measured.

A separate run of F20R was to analyze the cooling system at steady state, and in this case, the temperature of the radiator was measured. All the average values used are as follows.

$$T_{air_{in}} = 90\text{ }^{\circ}\text{F (ambient air temperature)}$$

$$T_{air_{out}} = 189\text{ }^{\circ}\text{F}$$

$$T_{rad_{in}} = 223\text{ }^{\circ}\text{F}$$

$$T_{rad_{out}} = 184\text{ }^{\circ}\text{F}$$

$$P_{engine} = 20\text{ HP}$$

$$\omega_{engine} = 8000\text{ RPM}$$

The analysis model will be based on the NTU-effectiveness method, using air-side heat transfer coefficients derived by Kays & London's wavy fin data [3]. This model will be validated against the MoTeC race data and other F20R data runs. Please see Appendix D for the NTU method approach specific to this automotive radiator analysis.

The cooling system will reject a certain amount of the coolant to the ambient air through the radiator core. At steady state, the heat rejected by the radiator needs to equal the amount of heat added by the engine. It can be assumed that approximately 30% of the power produced by the engine will be rejected to the coolant. Any remaining heat is lost through exhaust and friction heat. Using the analysis done in Section II, the radiator must reject the following:

$$Q_{rad} = Q_{engine}$$

$$Q_{engine} = 57,000 \frac{BTU}{hr} = 16,707\text{ Watts}$$

However, it is important to note that the coolant temperature entering the radiator is not fixed but is set by the balance between engine heat addition and the coolant's ability to absorb and transport the heat. If the mass flow rate of the coolant drops, the coolant will spend more time in the engine, thus picking up more heat. Though the coolant has more time to dissipate heat through the radiator core. This is why the temperature of the coolant going into the radiator is assumed to be constant to analyze the radiator's performance. The radiators share the same core geometry, including the tube height, fin pitch, fin thickness, fins per inch (FPI), and core depth. The total number of tubes does vary, however. Table I shows the geometries that were used for this analysis.

Table I - Basic Radiator Geometry Parameters

Parameter	F20R Radiator	F26 Radiator
Width x Height	12" x 9"	9" x 6"
Frontal core face area (A_{face})	108 in ²	108 in ² (2*54 in ²)
Core depth (D_{core})	1.25 in	1.25 in
Total tubes	24	16
Number of passes	2	2
Tubes per pass	12	8
Fins per inch (FPI)	15	16
Fin thickness (t_{fin})	0.1 mm	0.1 mm

To analyze the two-radiator configuration in series, it must be understood that the coolant runs through two cores sequentially, each with a fresh air stream. This is thermodynamically favorable because each core sees fresh cool air, rather than potentially pre-heated air from a preceding stage. For the analysis, the temperature at the exit of the first radiator will be used as the $T_{in,2}$ of the second radiator. However, the cost of increased plumbing complexity for the coolant tubes and pressure drop may be of worry.

Another consideration is the air inlet temperature. The model uses T_{atm} to be 80-90 °F, since that was measured on the MoTeC data for the ambient conditions on race day. However, the actual air temperature entering the core inside the sidepod is likely higher than ambient for several reasons. Track surface radiation for the asphalt and rubberized

concrete may make the air closer to the ground hotter. Sidepod heat soak from the engine, exhaust, and drivetrain could potentially be radiating heat into the sidepod enclosure. The rationale behind this concern is due to the extreme temperature measurements located at the back of the temperature fan. This may not be totally representative of the air outlet temperature as a whole and could be extremely concentrated or affected by external factors. The coolant does not travel through the core at a uniform temperature. Fins and tube walls near the coolant inlet are significantly hotter than those near the outlet, so air passing over these hot inlet-side fins may be a reason for such high temperature readings if the thermocouple happens to be biased in positioning towards the inlet. Instrumenting a thermocouple near the actual radiator face inlet during a run would provide a more accurate input than simply using the ambient temperature. Assuming the ambient temperature provides an optimistic analysis of the heat rejection capacity of the air. This spatial behavior of the radiator does present a fundamental limitation of the NTU single-node model. A multi-node model would be required if resolving the local distribution was necessary, but this is beyond the scope of this analysis, since a more global approach gives us a better overall understanding of the radiator. The math outlined in Appendix E is essential for the simulation of both radiators. Simulation in Fig. 1 (see Appendix E for MATLAB script) shows that the F20R radiator may not have been seeing any ram air at all to achieve such temperatures, but the fan would provide some air velocity if it were turned on. It makes more sense that the thermocouple was potentially facing the issues mentioned prior.

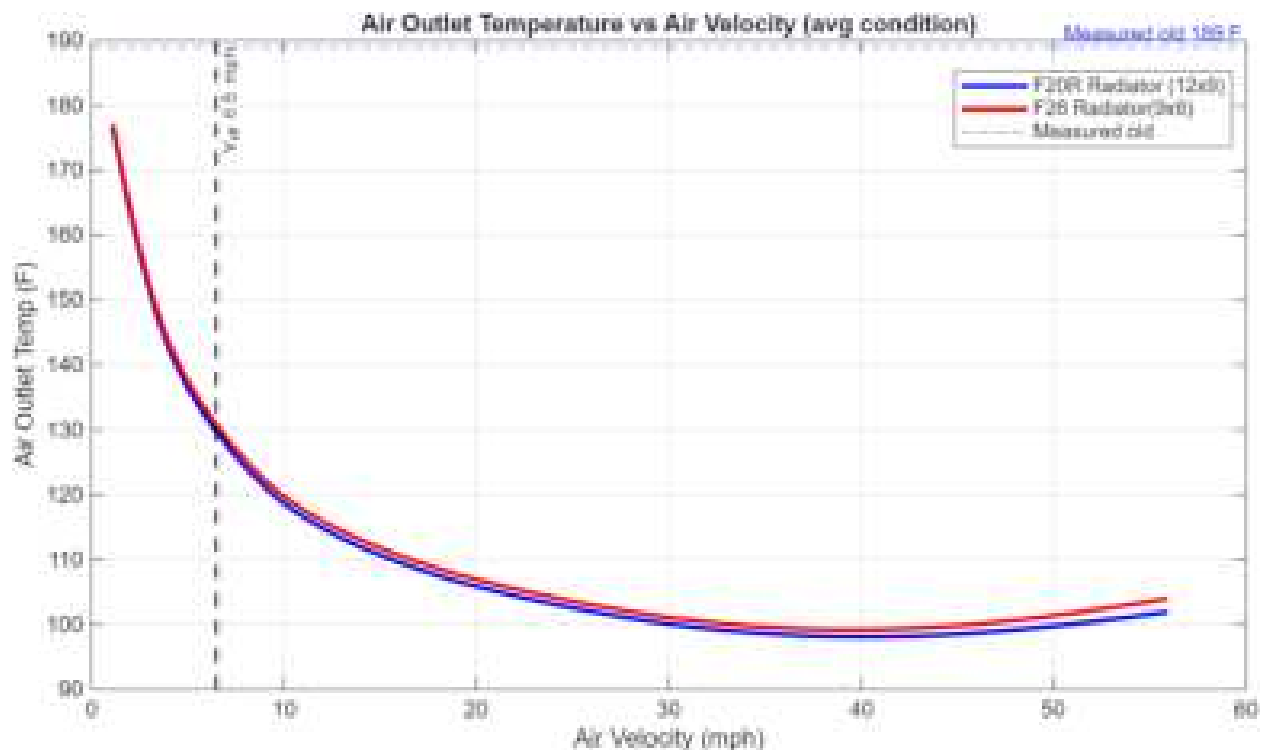


Figure 1. Air Outlet Temperature vs. Air Velocity

Using the MATLAB script outlined in Appendix E, the coolant outlet temperature can be plotted as either a function of air velocity or mass flow rate. Thus, we can get a better comparison of the difference between radiators. Note that the mass flow rate was assumed at the average RPM (8000 RPM) for the endurance run data when plotting the

coolant outlet temperature versus air velocity. The graph used for this approximation is shown in Fig. 2 from Ramin's report [2].

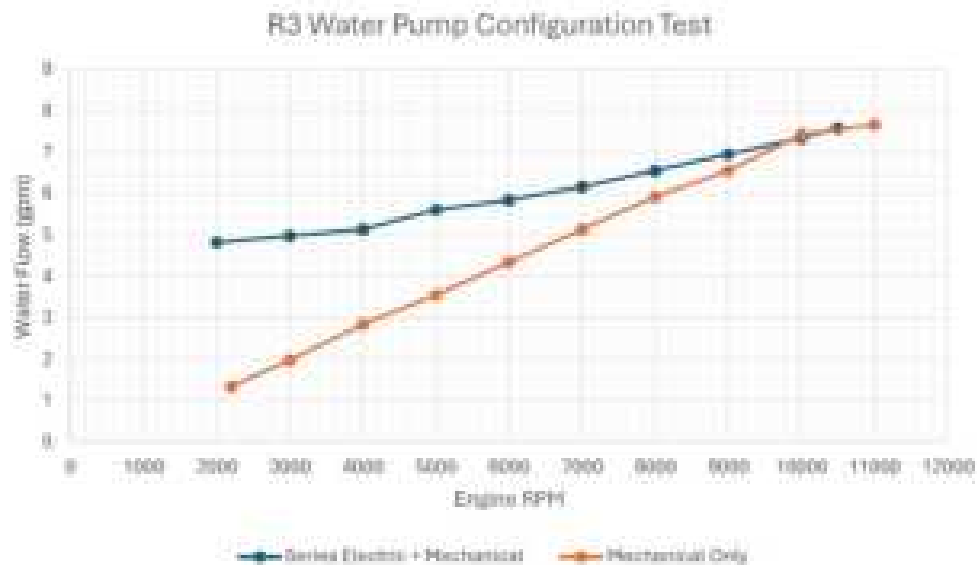


Figure 2. Water Pump Test [2]

Plotting the coolant outlet temperature versus the mass flow rate (mass flow rate provided at the pump, not through the core tubes) at the effective velocity of 6.5 mph demonstrates how heat capacity plays a small role in cooling. The simulation using the NTU method shows a drop of $\sim 8^\circ\text{F}$ at the current mass flow rate, indicated by the vertical dashed line on the graph in Fig. 3. Lowering the mass flow rate by around 0.1 kg/s could result in $\sim 10^\circ\text{F}$ drop from the coolant inlet temperature while maintaining operation in the turbulent region. Intuitively, this appears to be a conservative analysis, but the model must be compared to the actual data. The NTU method may be underpredicting the cooling of the radiator, but the limited data also affects the true validity of the simulation. The physical run of F20R, where the coolant outlet temperature was measured (engine coolant out temperature was measured as well), was $\sim 184^\circ\text{F}$ at steady state (heat soaked). The engine outlet temperature was $\sim 195^\circ\text{F}$ and can be assumed to be the radiator inlet temperature.

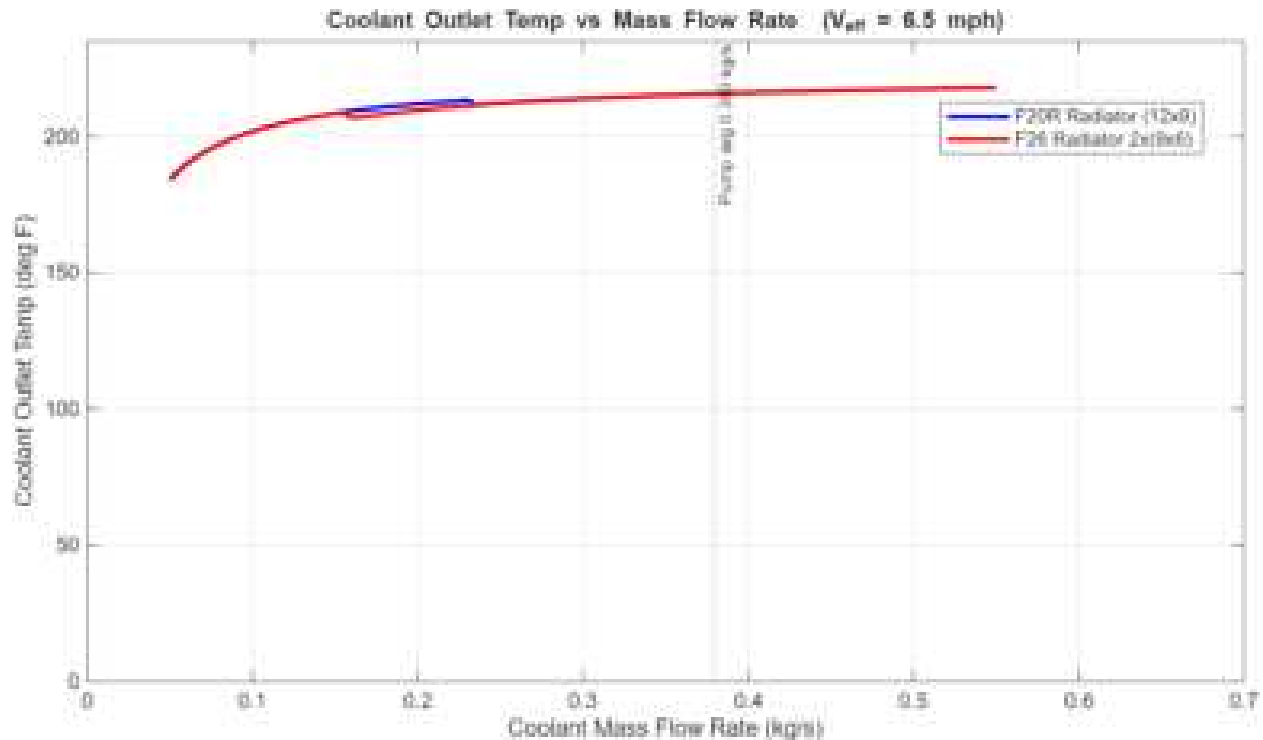


Figure 3. Coolant Outlet Temperature vs. Mass Flow Rate

Running the simulation using the endurance data, the F20R and F26 lines are essentially right on top of each other, showing that both offer the same amount of cooling due to the almost equivalent core frontal areas. The two-radiator configuration for F26 will reach turbulence prior to the F20R radiator, due to the increased mass flow rate, providing a potential benefit of increased cooling at lower RPMs. A similar trend with the coolant outlet temperature and air velocity can be observed. This shows the potential conservative approach of the NTU method but ultimately highlights the difference between radiator performance. In this case, the difference is essentially negligible. Letting more air into the sidepods could offer a slight cooling benefit to decrease coolant outlet temp a few degrees Fahrenheit, but as mentioned before, there might be a diminishing return on fin effectiveness.

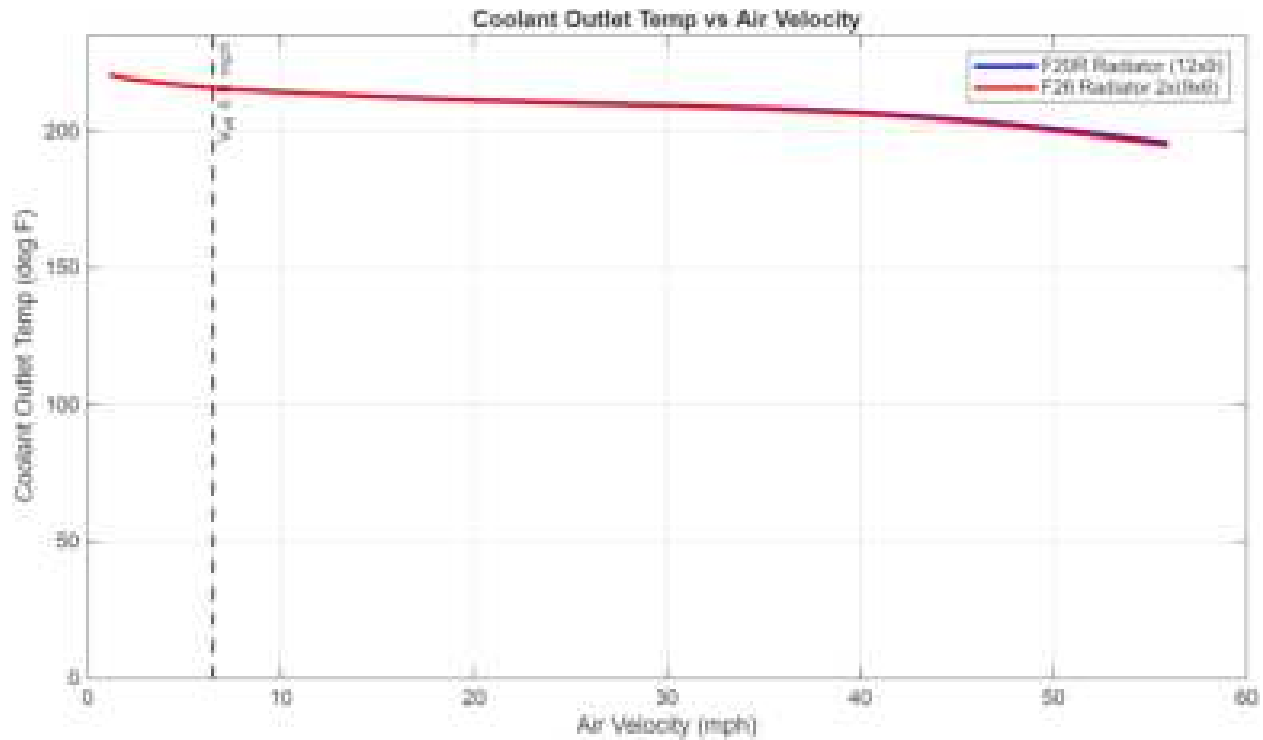


Figure 4. Coolant Outlet Temperature vs. Air Velocity

Now, the simulation model needs to be compared to the actual data as mentioned before, using engine coolant outlet temperature as our radiator inlet and the coolant outlet temperature ($T_{rad,in} = 195^{\circ}\text{F}$, $T_{rad,out} = 184^{\circ}\text{F}$). The outlet temperature at the average mass flow rate then will be compared to the physical data, given an input of 195 F. This temperature may also drop 1-2 F as it makes its way through tubing and radiator inlet. See Fig. 5.

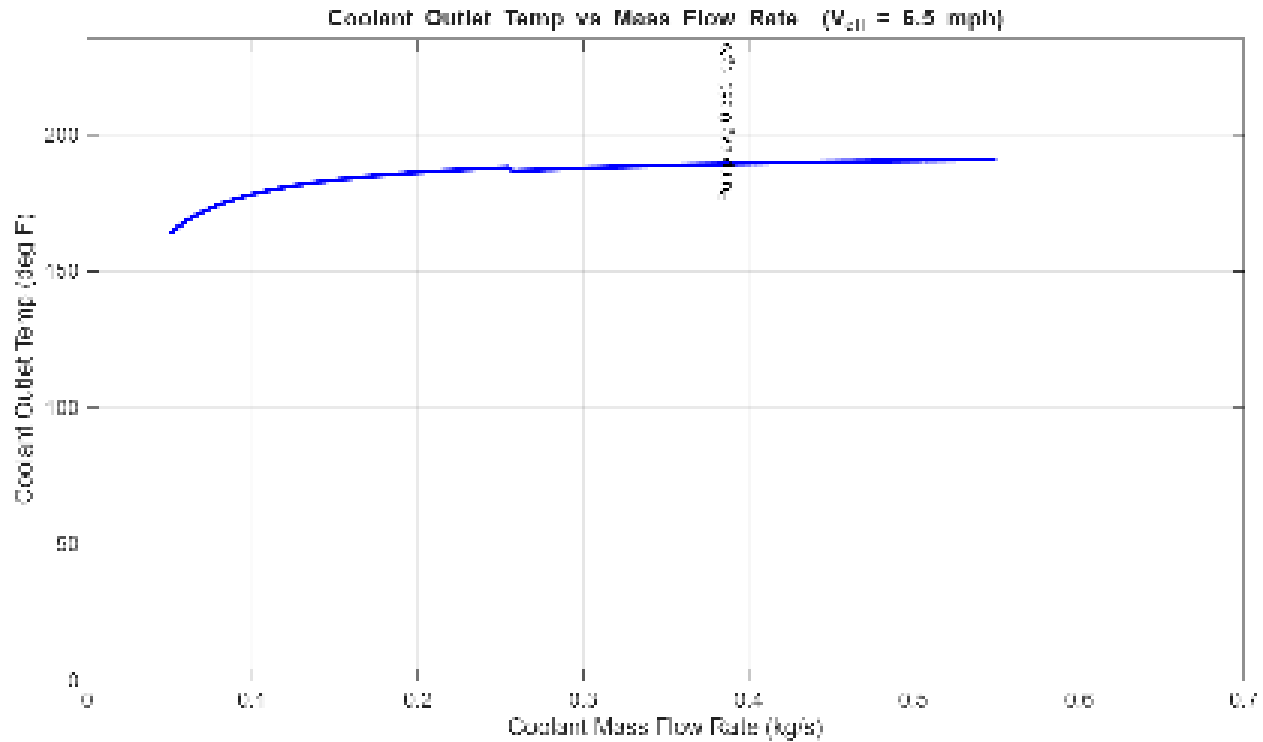


Figure 5. Coolant Outlet Temperature vs. Mass Flow Rate using F20R heat soak data input (195 °F)

The NTU model predicts that the outlet temperature of the coolant will be approximately 185-186.5 °F, depending on how the mass flow rate varies before it reaches the core. This is relatively close to the 184 °F, therefore proving the validity of the NTU model, while maintaining a slightly conservative approach. Due to the lower temperatures of the inlet temperature endurance data, the average RPM for the run could have been lower as well, which would potentially decrease the coolant outlet temp, further proving the NTU model's accuracy. Other parameters, such as the ambient temperature of the air for that day and wind velocity were not recorded. This model also does not account for any conduction or convection that may have occurred between coolant piping before entering the radiator core, since this analysis only accounts for the heat transfer that occurs through the core. Therefore, the actual temperature difference is most likely even closer to the track data. Overall, the NTU method proves to be a good tool to compare radiator performance through their geometries, mass flow rates, and number of tubes. The current F26 configuration is the better choice between the two, since F26 offers better packaging in smaller sidepods and even cooling. Smaller sidepods are beneficial to the aero package, and even cooling can potentially offer better heat exchanger performance, reducing the need for as much fan power. Although, it may prove more useful to analyze the heat rejected by the radiator in comparison to the required heat rejection due to the assumed 30% heat sent to the cooling system. This will need to be compared to the 16,707 W calculated earlier for Q_{required} . For F20R:

$$Q_{\text{cap}} = \varepsilon C_{\text{min}}(T_{\text{rad,in}} - T_{\text{atm}}) = 6421 \text{ W}$$

For F26:

$$Q_{\text{cap}} = \varepsilon C_{\text{min}}(T_{\text{rad,in}} - T_{\text{atm}}) = 6481 \text{ W}$$

F20R and F26 reject about 38% of the required heat needed to keep the system stable. Thus, both radiators prove to be ineffective in terms of the heat that needs to be removed from the cooling system to maintain the temperature going into the system. These values are purely theoretical based on the NTU method. If we look at the actual temperatures measured by the system from the F20R heat soaked run in Ramin's report [2], using engine coolant outlet temperature as our radiator inlet and the coolant outlet temperature ($T_{\text{rad,in}} = 195 \text{ °F}$, $T_{\text{rad,out}} = 184 \text{ °F}$), the system proves to be rejecting far less than the required, again.

$$Q_{actual} = \varepsilon C_{min}(T_{rad,in} - T_{rad,out}) = 5156 \text{ W} \left(17,593 \frac{\text{BTU}}{\text{hr}} \right)$$

If the concern is simply to achieve a lower coolant outlet temperature, then a radiator with larger core frontal area with more tubes is recommended. Here is a proposed radiator that uses a 18" length by 12" wide core, using the same 1.25" core depth. This radiator uses 32 tubes to slow down the mass flow rate through each tube but still maintains turbulence at the average pump mass flow rate.

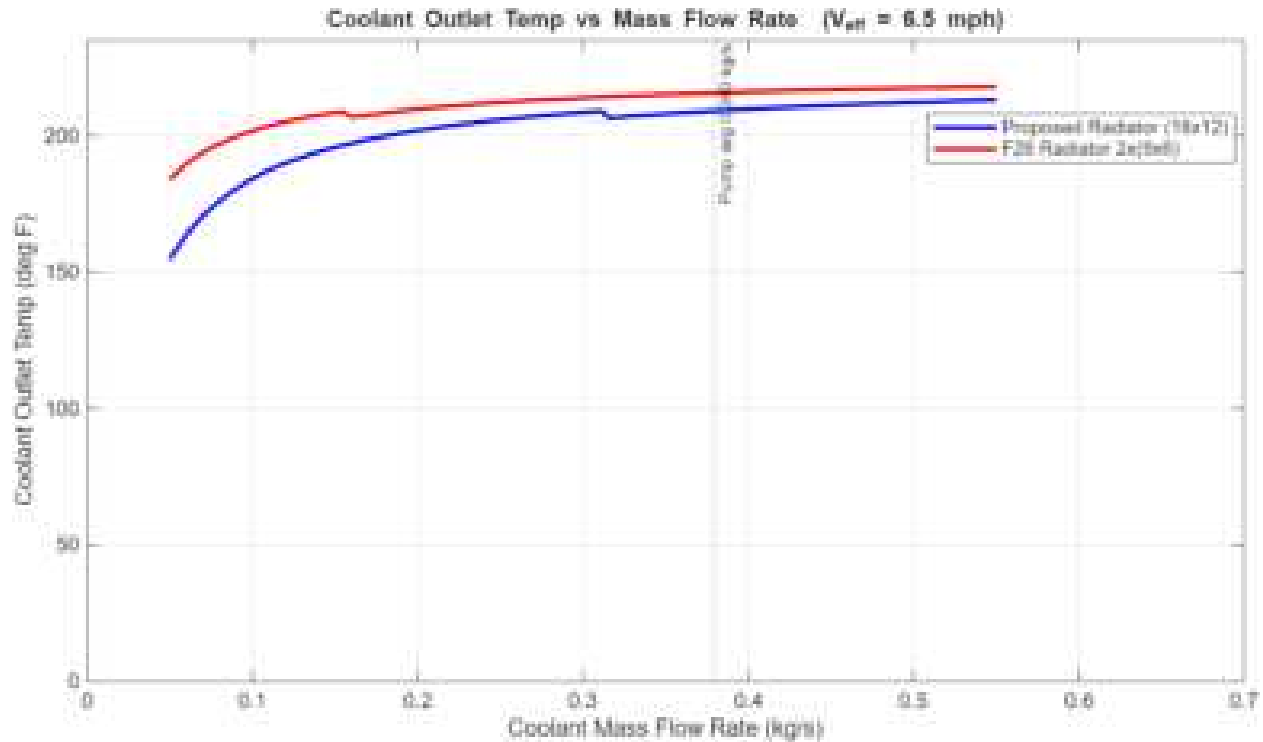


Figure 6. Proposed Radiator Solution (18"X12", 32 total tubes)

This solution is a quite large radiator but demonstrates that to lower the outlet temperature, the core frontal area must be increased. This is better for airflow through the radiator and providing a larger amount of surface area for the heat exchanger to remove heat. This solution only rejects about 11,000 W, still short of required heat needed to be rejected, but much higher than the F20R and F26 radiators.

However, it is necessary to further test the F20R and F26 radiators to determine a more accurate conclusion for the coolant outlet temperature and actual heat rejection. Current data shows the heat rejection is slightly higher than theoretical analysis. A full endurance run that captures all temperatures, such as temperatures of the coolant and air both directly before and after the radiator. The air velocity using rake data would also prove useful. Building a model from flow bench and race data would most likely provide a more accurate analysis approach, though the NTU model does show promise on its heat transfer prediction.

IV. Ansys Fluent Analysis

Before beginning the Ansys Fluent analysis, the radiator placement and surrounding vehicle geometry must be considered because they influence the ambient airflow conditions. The UTA FSAE team packages the radiators inside the sidepods, which help direct airflow toward the cooling system while also protecting the radiators from debris. The sidepods also contribute to the vehicle's aerodynamic behavior by influencing how air flows around the car. However, due to the nature of the large front wings on FSAE cars, it was initially assumed that only a small amount of ram air would enter the sidepods and pass through the radiator. Under this assumption, fan selection and fan sizing would have a greater effect on cooling performance than the exact radiator orientation within the sidepod.

Other highly competitive FSAE teams, such as Texas A&M (placed 3rd for IC in 2025), do not utilize a sidepod configuration but rather mount the radiator in the back of the car next to the engine, as shown in Fig. 7. It is also tilted backwards, assumed to be for more surface area while still mitigating the frontal area. This is important when considering the aerodynamic forces, and the removal of sidepods may also reduce this area and remove mass from the entire car system. The sidepods will be more exposed to the open air, which could aid in cooling. It is important to look at the CFD analysis to determine how much air is flowing into the sidepods.



Figure 7. Texas A&M FSAE without sidepods [6]

Looking at a cross-sectional front view of the side pods right as the air begins to flow over, a large amount of pressure towards the top of the side pod begins to enter. Thus, proving the initial belief wrong that there is no ram air going into the sidepod. The low amount of pressure at the bottom is due to the front wings guiding the air over the top of the car, as well as the tires blocking most of the air that would be going into the side pods. These cross sections are shown in Figs. 8 and 9, respectively. However, even with a nozzle effect at the inlet, which would cause a pressure drop, the pressure coefficient is still greater than 1 going into the sidepod, again confirming the existence of ram-air. A deeper analysis using Ansys will need to be conducted to better determine the heat transfer effects on the radiator from the ram-air in comparison to how much work the fan does.



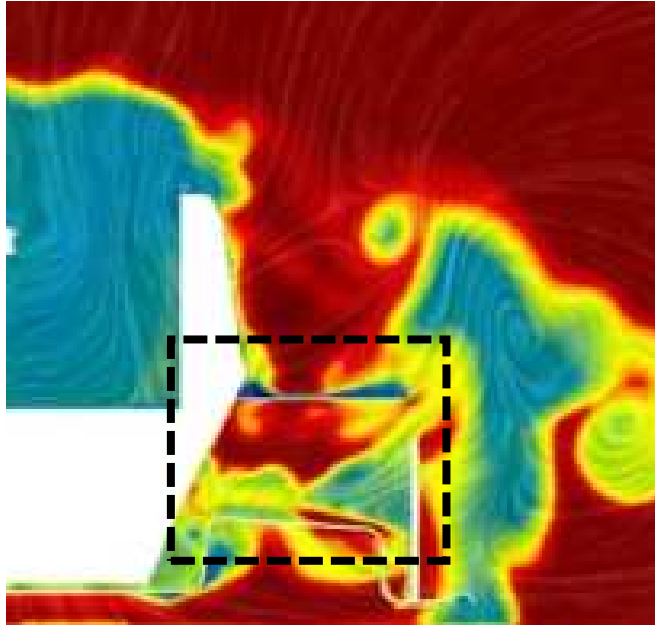


Figure 8. CFD Front View Cross-Section Sidepod Inlet

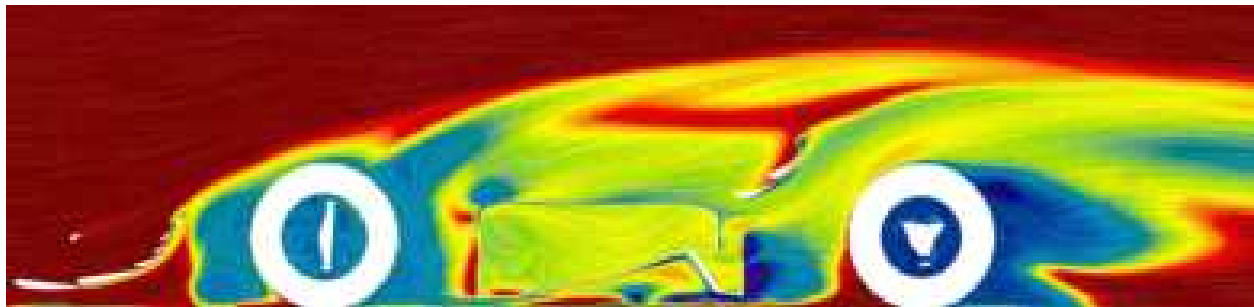


Figure 9. CFD Side View Cross-Section at Sidepod

With the sidepods justified, the system can then move on to the modeling of the radiator. The radiator was modeled based on the calculations gathered during the numerical analysis. Proposed radiator core dimensions for a single radiator configuration to match the efficiency of the F26 dual radiator configuration are as follows: length of 14.8 in, height of 9.8 in, and width of 1.25 in. A full rendered model of the proposed radiator can be seen below.

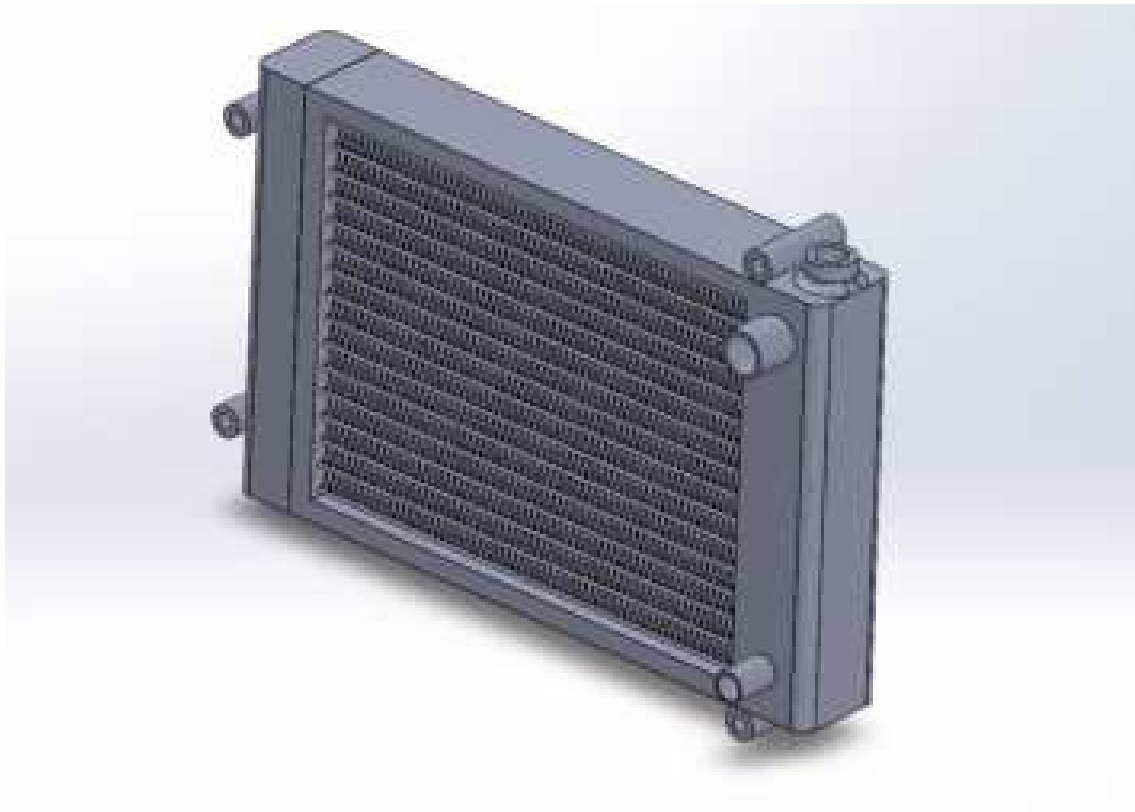


Figure 10. Radiator Model



Figure 11. Sectioned View of Radiator

Next, we will move onto Ansys Fluent simulation to confirm the theory behind the calculations. Some simplifications need to be made in order to gather usable data while saving computational costs. The radiator will be simplified to contain the top 6 tubes and 5 sets of fins, and only the first pass of the radiator. With this setup, the crossflow will be omitted due to the complexity and time shortages; however, the model can easily conform to a crossflow setup and can be re-examined during future work. For this simulation, emphasis on the convection of the coolant to the aluminum, and the aluminum to the ambient air, will be placed. The meshing process for a fluid conjugate heat transfer is vital to the quality of the resulting simulation; thus, a mesh study was done to improve results. Findings show that a mixture of body sizing and inflation layers can be used to yield a quality mesh. A body sizing of 6 mm is placed on all the fins, while the aluminum tubes have a body sizing of 10 mm. The most important mesh setting for fluid flow is to place inflation layers in internal geometry of the tube; this helps with the calculations of interactions between the coolant and aluminum. The downside to this precise internal mesh of the coolant is that it creates a section of the inlet and outlet that has more nodes than the rest of the face. This creates an artifact in the results but knowing this will help inform deductions. The speed of the coolant going into the pipes is approximately 0.17 meters per second or 0.38 miles per hour, which is based on 7 gallons per minute being supplied by the water pump. The heat transfer from the coolant to the piping is $8635 \frac{W}{m^2 \cdot K}$, then the aluminum will have a transfer rate $52 \frac{W}{m^2 \cdot K}$ to the air; however, do note that the air is static and will affect the results. The coolant in the piping was set to have a turbulence ratio of 0.05 due to the Reynolds' number found from the numerical results at approximately 11500. When running the simulation, the number of iterations was set at 50, and the loosely coupled conjugate heat transfer was turned on, set to 50 iterations as well. The system was set to measure the outlet temperature of each pipe and record it. Once the simulation begins, the initial temperature for each outlet is a dramatic drop, which then will increase a little and then stabilize; however, for some pipes, there will be a section that will experience some fluctuations before truly stabilizing. The results from this Ansys Fluent study can be seen below in the following figures.

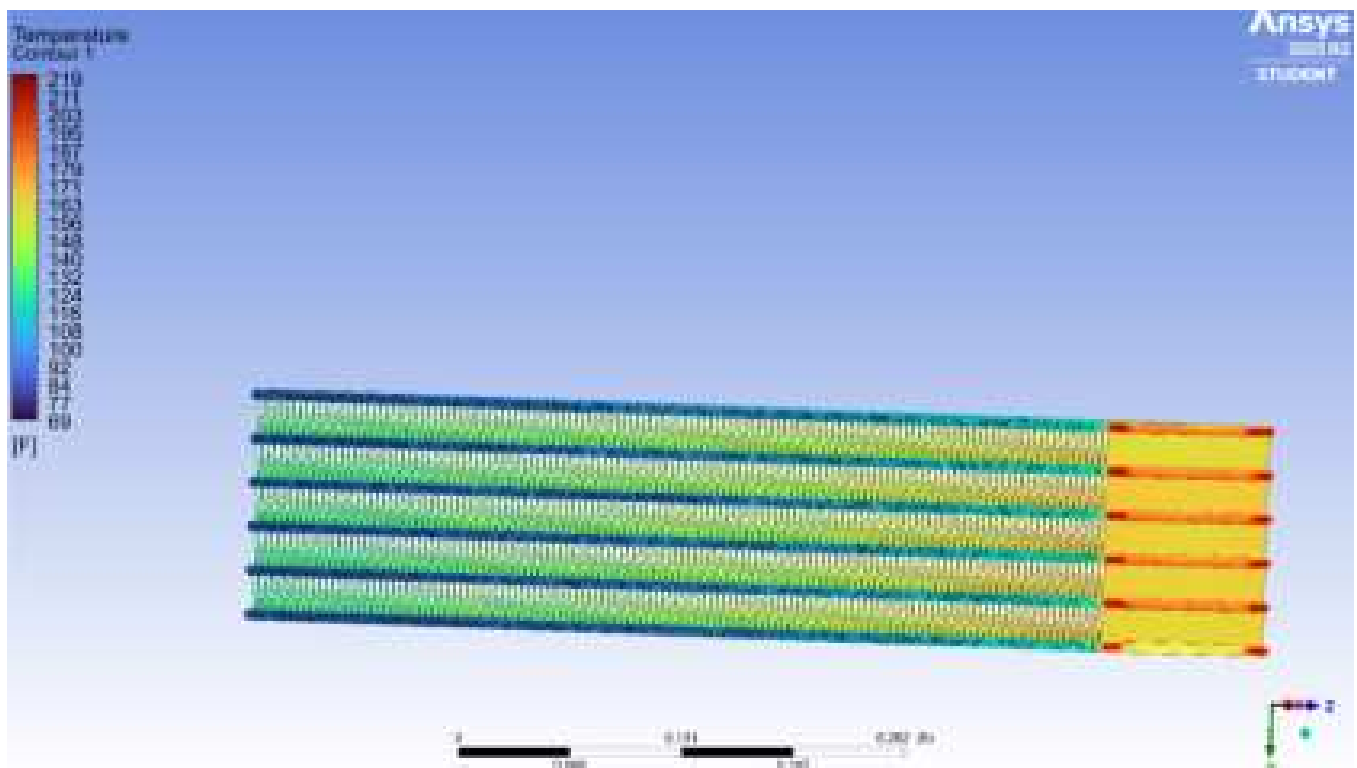


Figure 12. Final Radiator Orthogonal View

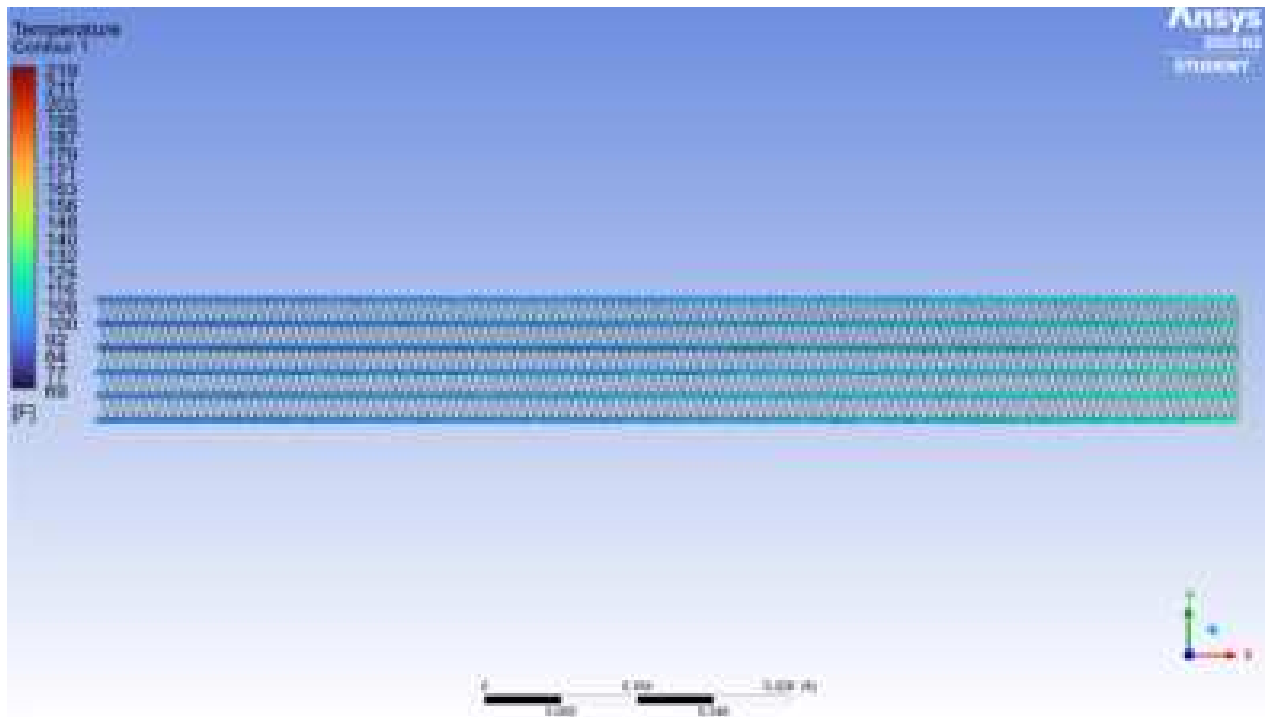


Figure 13. Final Radiator Side View

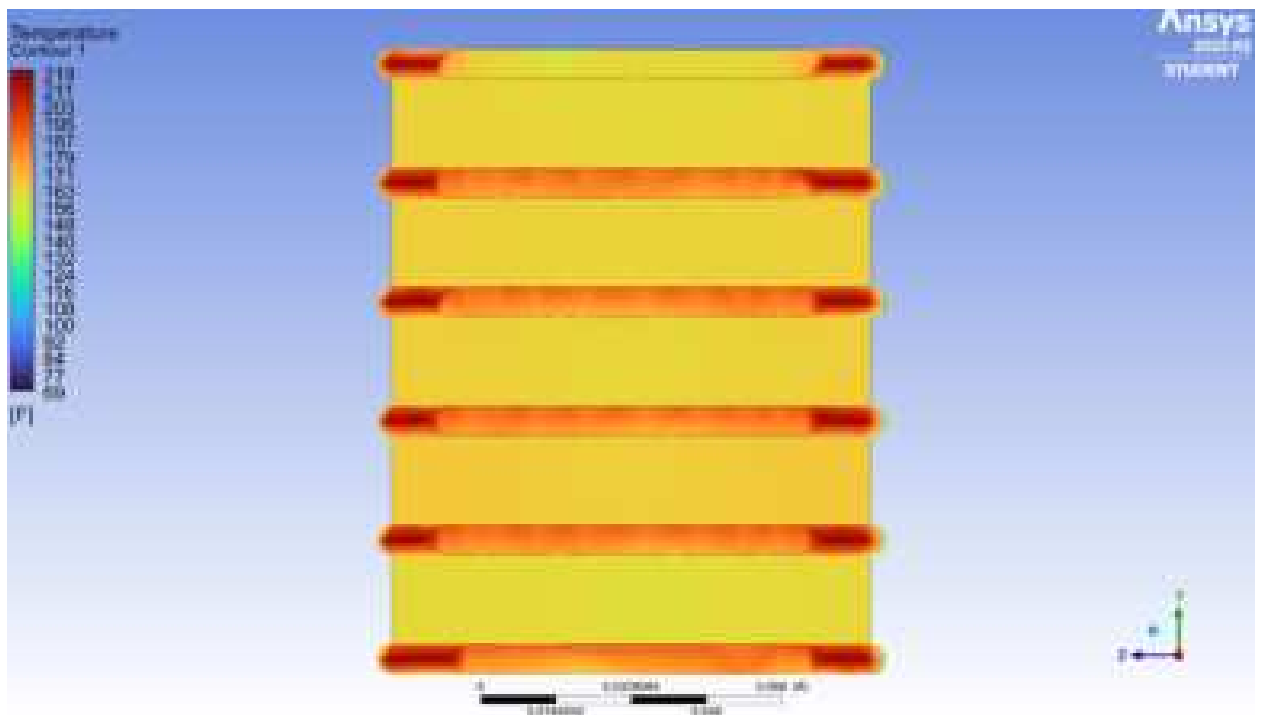


Figure 14. Final Radiator Inlet View

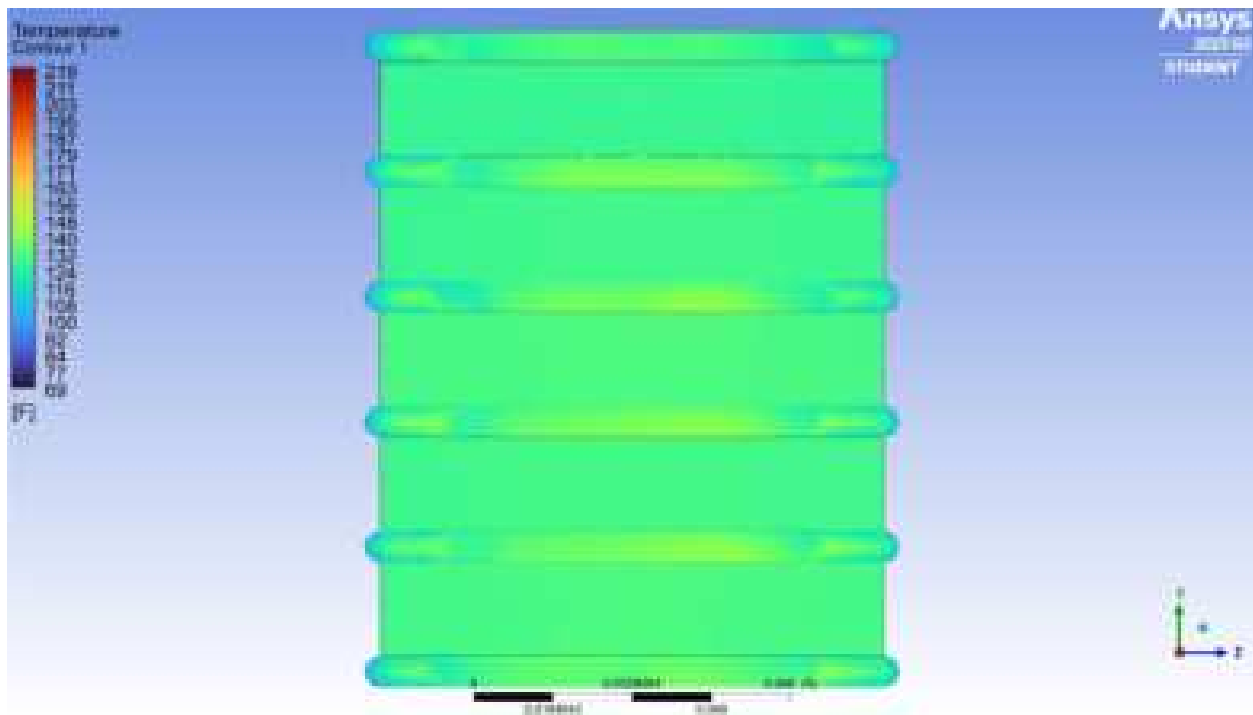


Figure 15. Final Radiator Outlet View

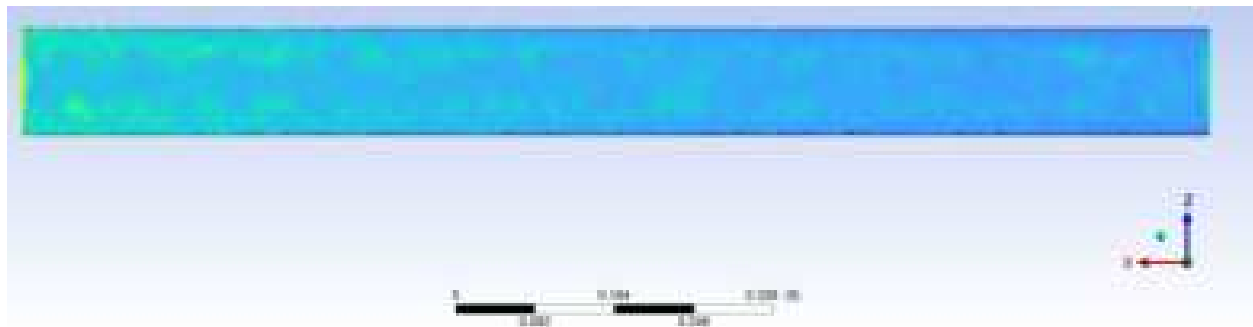


Figure 16. Final Radiator Top-Down View (Inlet to the left)

After reviewing the gradients collected from the simulation, it is clear to see that the model is cooling faster than predicted through the numerical analysis. This is likely due to the defect of localized hot and cool regions within the inlets and outlets. The localized hot regions near the tube corners are likely caused by inlet-development and geometry/mesh effects. The coolant enters at the prescribed temperature, and the sharp/rounded tube corner regions create strong local temperature gradients where the flow first contacts the tube wall. Knowing where this defect comes from, the analysis should rely on downstream temperature trends, area-averaged outlet temperatures, and total heat-transfer rates rather than local extrema near the inlet. With this principle in mind, it can be ascertained that the tubes are experiencing a uniform heat flux to the surrounding area as the coolant moves downstream. This shows that any future model representing this system can be simplified down to 1-3 tubes and be extrapolated out for the entire radiator. This reopens the possibility of doing a crossflow analysis and should be considered for future work.

V. Future Tools for Sizing

The radiator sizing tool was created to convert user-defined inputs into thermal, fluid, geometric, and pressure outputs that help the user determine whether a selected radiator geometry is viable for the conditions it will be subjected to. It is organized into five sections: Inputs, Outputs, Checks, Unit Conversion, and Information, with the main calculations happening between the Inputs, Outputs, and Checks sections and Unit Conversion and Information being used to aid the engineer in the process of sizing their radiator.

All equations used to develop the radiator sizing tool, unless otherwise stated, are from Fundamentals of Heat and Mass Transfer [7]. The tool utilizes the effectiveness-NTU (Number of Transfer Units) method for heat exchangers, which utilizes the maximum possible heat transfer rate that the system can experience and compares it to the actual heat transfer rate (input by the user), as seen below:

$$q_{max} = C_{min}(T_{h,i} - T_{c,i})$$

$$\varepsilon = \frac{q}{q_{max}}$$

This effectiveness is then used in conjunction with the heat capacity ratio to determine the NTU using the relationship for crossflow with both fluids unmixed, as seen below:

$$\varepsilon = 1 - \exp\left[\left(\frac{1}{C_r}\right)(NTU)^{0.22}\{\exp[-C_r(NTU)^{0.78}] - 1\}\right]$$

$$C_r = \frac{C_{min}}{C_{max}}$$

Note that the tool was created in Excel, so, in the essence of keeping it simple, the NTU is calculated using an outside MATLAB code, as seen in Fig. 17, and input into the tool manually.

```

1  clc
2  clear
3  format compact
4
5  % inputs
6  epsilon = 0.33;
7  Cr = 0.18;
8
9  % Check Inputs
10 if epsilon <= 0 || epsilon >= 1
11     error('Effectiveness must be between 0 and 1.')
12 end
13
14 if Cr <= 0 || Cr > 1
15     error('Cr must be between 0 and 1.')
16 end
17
18 % Both fluids unmixed
19 fun = @(NTU) 1 - exp((1/Cr)*(NTU.^0.22).*(exp(-Cr*(NTU.^0.78)) - 1)) - epsilon;
20
21 % Solve for NTU
22 NTU_guess = 1;
23 NTU = fzero(fun, NTU_guess);
24
25 % Output
26 disp(['NTU = ', num2str(NTU)])

```

Figure 17. MATLAB NTU Calculator

The user input process is split up into four sections: Thermal, Fluid, Geometric, and Pressure. The Thermal section outlines the required operating conditions, and as seen in Fig. 18, the user can define:

1. Heat Rejection Rate (which can be found using the method outlined in Appendix A)
2. Air Inlet Temperature (corresponds to ambient air temperature)
3. Coolant Inlet Temperature (corresponds to engine operating temperature)
4. Specific Heats (found in property tables)

Thermal Inputs					
Shared					
Parameter		Value		Unit	
Heat Rejection Rate, Q		6		kW	
Air			Coolant		
Parameter	Value	Unit	Parameter	Value	Unit
Inlet Temp, $T_{c,in}$	35	°C	Inlet Temp, $T_{h,in}$	95	°C
Specific Heat, $C_{p,c}$	1.005	kJ/kg·K	Specific Heat, $C_{p,h}$	4.214	kJ/kg·K

Figure 18. Thermal Inputs

The Fluid Properties section defines the values used in the calculation of Reynolds and Nusselt numbers later, and as seen in Fig. 19, the user can define:

1. Fouling Factors (used to make design conservative)
2. Densities
3. Dynamic Viscosities
4. Conductivities
5. Prandtl Numbers
6. Mass Flow Rates

These fluid properties will stay constant between radiator sizes as they rely either on the engineer's preferences, in the case of the fouling factors, or the fluid's temperature, which is defined by the requirements of the radiator.

Fluid Property Inputs					
Air			Coolant		
Parameter	Value	Unit	Parameter	Value	Unit
Fouling Factor, $R_{f,c}$	0.00001	$m^2 \cdot K/W$	Fouling Factor, $R_{f,h}$	0.0001	$m^2 \cdot K/W$
Density, ρ_c	1.18	kg/m^3	Density, ρ_h	960.61	kg/m^3
Dynamic Viscosity, μ_c	0.0000186	$Pa \cdot s$	Dynamic Viscosity, μ_h	0.000289	$Pa \cdot s$
Conductivity, k_c	0.028	$W/m \cdot K$	Conductivity, k_h	0.679	$W/m \cdot K$
Prandtl, Pr_c	0.707	-	Prandtl, Pr_h	1.8	-
Mass Flow Rate, m_{dot_c}	0.3	kg/s	Mass Flow Rate, m_{dot_h}	0.4	kg/s

Figure 19. Fluid Property Inputs

The Geometric section is self-explanatory, and as seen in Fig. 20, the user can define:

1. Core Width, Height, and Depth (constrained by the space available for the radiator)
2. Tube Height and Wall Thickness
3. Number of Passes and Tubes/Pass
4. Tube Conductivity

5. Free Flow Area Fraction
6. Fin Thickness, Length, and Fins/mm

While some of the geometric properties, such as the core dimensions, will stay constant between radiator comparisons, other inputs, such as the number of passes and tubes per pass, are critical inputs for the engineer.

Geometric Inputs		
Parameter	Value	Unit
Core Width, W	203.20	mm
Core Height, H	142.40	mm
Core Depth, D	31.75	mm
Tube Outer Height, $H_{t,o}$	2.00	mm
Tube Wall Thickness, t_w	0.36	mm
Number of Passes, N_{pass}	2	-
Tubes/Pass	8	-
Tube Conductivity, k_t	205	W/m·K
Free Flow Area Fraction, ϕ_{ff}	0.5	-
Fin Thickness, t_f	0.08	mm
Fin Length, L_f	7.92	mm
Fins per mm	0.591	fins/mm

Figure 20. Geometric Inputs

Finally, the Pressure section allows the user to define:

1. Allowable Pressure Drops
2. Minor Head Loss Coefficient (used to make the model conservative by accounting for geometric factors in the radiator)

Which can come from industry standards or be defined based on experiments conducted by the engineer. Fig. 21 shows the Pressure input section.

Pressure Inputs					
Air			Coolant		
Parameter	Value	Unit	Parameter	Value	Unit
Allowable Pressure Drop, ΔP_{allow}	100	Pa	Allowable Pressure Drop, ΔP_{allow}	10	kPa
			Minor Head Loss Coefficient, $K_{H,tot}$	3	-

Figure 21. Pressure Inputs

Upon entering all the required inputs, the tool automatically displays the Thermal, Geometric, Fluid, and Pressure outputs, with the process for each being described below.

The Thermal Section, seen in Fig. 22, calculates the heat capacitance for both the air and coolant, determines the minimum and maximum values, and calculates the heat capacity ratio. It then uses these values to calculate both the maximum heat transfer rate and the effectiveness, ultimately finding the NTU and the required UA. Equations for the heat capacitance of the air and the coolant, along with the equation for the required UA, are provided below.

$$C_c = \dot{m}_c c_{p,c}$$

$$C_h = \dot{m}_h c_{p,h}$$

$$UA_{req} = NTU * C_{min}$$

Thermal Outputs							
Shared							
Parameter			Value		Unit		
C _{min}			302		W/K		
C _{max}			1686		W/K		
Capacity Ratio, C _r			0.18		-		
Q _{max}			18.09		kW		
Effectiveness, ε			0.33		-		
Solved NTU			0.419		-		
Required UA			126.3285		W/K		
Air			Coolant				
Parameter		Value	Unit	Parameter		Value	Unit
Thermal Capacitance, C _c		302	W/K	Thermal Capacitance, C _h		1686	W/K
Outlet Temp, T _{c,out}		55	°C	Outlet Temp, T _{h,out}		91	°C

Figure 22. Thermal Outputs

The Geometric Section, seen in Fig. 23, then computes the different areas and perimeters associated with the geometry, along with other factors such as the fin pitch, total number of fins, and hydraulic diameters, which are used later in the calculation of the Reynolds numbers. The equations used by the Geometric section (not Bergman and Lavine) can be found in Appendix F. At this point, it can be noted that the Fluid and Pressure sections for the air side were calculated, assuming the airside problem as an internal pipe rather than using Colburn j-factor correlations. This was done for the purpose of keeping the tool automatic and keeping calculations consistent between the air and coolant sides.

Geometric Outputs					
Air			Coolant		
Parameter	Value	Unit	Parameter	Value	Unit
Outside Tube Surface Area, $A_{s,o}$	226879	mm ²	Total Tubes, T_{tubes}	16	-
Frontal Area, A_{front}	28936	mm ²	Single Tube Section Area, $A_{t,cross}$	41	mm ²
Free Flow Area, A_{ff}	14468	mm ²	Tube Outside Perimeter, P_o	70	mm
Fin Pitch, p_{fin}	1.69	mm	Tube Inside Perimeter, P_i	66	mm
Total Fins, T_{fins}	1801	-	Internal Tube Surface Area, $A_{s,i}$	214843	mm ²
Fin Area, A_f	905944	mm ²	Single Pass Flow Area, A_{flow}	328	mm ²
Total External Area, A_{total}	1132823	mm ²	Tube Hydraulic Diameter, $D_{h,h}$	2.48	mm
Core Hydraulic Diameter, $D_{h,c}$	3.22	mm			

Figure 23. Geometric Outputs

After computing all geometric outputs, the tool then calculates the parameters in the Fluid section, as seen in Fig. 24. These parameters include the fluid velocities, Reynolds numbers, Nusselt numbers, convective coefficients, and the fin parameter, fin efficiency, overall surface efficiency, and the available UA. The fin parameter, m , describes how quickly temperature is lost from the fin, with larger fin parameter values relating to lower efficiency. Equations for the air and coolant side fluid properties can be found in Appendix G, with the equations for the fin parameter, fin efficiency, overall surface efficiency, and available UA being provided below.

$$m = \sqrt{\frac{2h_c}{k_f t_f}}$$

$$\eta_f = \frac{\tanh(mL_f)}{mL_f}$$

$$\eta_o = 1 - \frac{A_f}{A_{total}}(1 - \eta_f)$$

$$\frac{1}{UA_{avail}} = \frac{1}{h_h A_{s,i}} + \frac{R_{f,h}}{A_{s,i}} + \frac{t_w}{k_t A_{s,i}} + \frac{R_{f,c}}{A_{s,o}} + \frac{1}{\eta_o h_c A_{s,o}}$$

Fluid Outputs					
Shared					
Parameter		Value		Unit	
m=sqrt[2*h _c /(k _f *t _f)]		168.06		1/m	
Fin Efficiency, η _f		0.65		-	
Overall Surface Efficiency, η _{overall}		0.72		-	
UA _{avail}		163.04		W/K	
Air			Coolant		
Parameter	Value	Unit	Parameter	Value	Unit
Velocity, V _c	17.57	m/s	Velocity, V _h	1.27	m/s
Re _c	3594	-	Re _h	10473	-
Nusselt Number, Nu _c	14.00	-	Nusselt Number, Nu _h	45.12	-
h _c	231.60	W/m ² ·K	h _h	12342	W/m ² ·k

Figure 24. Fluid Outputs

Finally, the Pressure section, seen in Fig. 25, computes the Darcy friction factors for both fluids, then, in the case of the air side, it computes both the mass velocity and the expected pressure loss, and in the case of the coolant side, it calculates the equivalent length, major and minor pressure losses, and the expected pressure loss for the system. Equations for the pressure output calculations can be found in Appendix H. Note that the equations used in the pressure calculations come from Fundamentals of Fluid Mechanics [8].

Pressure Outputs					
Air			Coolant		
Parameter	Value	Unit	Parameter	Value	Unit
Friction Factor, f_D	0.041	-	Friction Factor, f_D	0.031	-
Mass Velocity, G_a	20.74	kg/m ² s	$L_{eq,H}$	406.4	mm
Total Head Loss, ΔP_{tot}	73.31	Pa	Major Head Loss, ΔP_{major}	3.96	kPa
			Minor Head Loss, ΔP_{minor}	2.32	kPa
			Total Head Loss, ΔP_{tot}	6.28	kPa

Figure 25. Pressure Outputs

The tool also provides multiple checks, as seen in Fig. 26, utilizing Excel's conditional formatting to give the engineer either a bright green or red visual to indicate whether the current parameters fall into the acceptable ranges set forth by the engineer. The Unit Conversion and Information sections, which were added to make using the tool easier, can be found in Appendix I.

Q_{max}	Effectiveness	UA_{avail}	ΔP_{air}	ΔP_{cool}
Qmax Ok	Effectiveness Ok	UA Ok	ΔP_{air} Ok	ΔP_{cool} Ok

Figure 26. Tool Checks

VI. Conclusion

This report evaluated the radiator cooling system for the UTA FSAE F26 racecar to determine whether the current dual-radiator configuration is necessary or if a single-radiator configuration could provide similar cooling performance. The analysis began with a thermodynamic estimate of the engine heat rejection requirement using E85 fuel properties, peak engine data, and average endurance operating conditions. Based on the estimated E85 lower heating value and fuel energy input, the peak coolant heat rejection was scaled to the average endurance power condition of 20 HP. This resulted in an estimated cooling requirement of approximately 57,000 Btu/hr, or 16,707 W, for the complete cooling system. For the current two-radiator configuration, this corresponds to approximately 29,000 Btu/hr per radiator under the average endurance design condition.

The analytical radiator comparison showed that the F20R single-radiator configuration and the F26 dual-radiator configuration have nearly the same total frontal core area, resulting in similar predicted cooling performance. The NTU-effectiveness model predicted outlet temperatures that were reasonably close to the available heat-soak data, with the model predicting approximately 189°F compared to the measured coolant outlet temperature of approximately 184°F. This suggests that the NTU model is useful for comparing radiator geometry and identifying general performance trends, even though it remains conservative and limited by the available experimental data.

The results also showed that both the F20R and F26 radiator configurations fall short of the estimated required heat rejection when evaluated strictly through the theoretical NTU method. The calculated heat rejection capacities were approximately 6421 W for the F20R radiator and 6481 W for the F26 radiator, which is only about 38% of the estimated required heat rejection. This indicates that the model may underpredict actual radiator performance, that the assumptions may be overly conservative, or that additional experimental data is needed to more accurately define the true heat load and airflow conditions.

The Ansys Fluent analysis provided a simplified conjugate heat transfer model of a representative radiator section. The simulation was useful for visualizing coolant-to-wall heat transfer behavior and identifying local temperature gradients, but it did not fully capture the complete crossflow behavior of the radiator because of simplifications made to reduce computational cost. The Fluent results showed faster cooling than predicted by the analytical model, but still remained close enough to support the general trends of the numerical analysis.

Overall, the current F26 dual-radiator configuration remains the preferred design based on the information currently available. While the F20R single-radiator configuration provides similar predicted thermal performance due to its comparable frontal area, the F26 layout offers packaging advantages, smaller sidepods, and more evenly distributed cooling. However, the final conclusion should not be based on the model alone. More complete testing is needed, including coolant inlet and outlet temperatures, air inlet and outlet temperatures, radiator face velocity, pump flow rate, and pressure drop across the system during a full endurance run. With this data, the NTU model, Fluent simulation, and Excel sizing tool can be validated and refined to determine whether a simplified single-radiator configuration is truly viable for future FSAE designs.

Appendix A – In-depth Numerical Analysis

The following process is the thermodynamic framework used to determine the required heat rejection capacity of the dual radiators for the F26 racecar. The objective is to determine the rate of heat that must be removed by the cooling system to maintain a steady engine operating temperature under peak load conditions.

The turbocharged Yamaha R3 engine receives energy from E85 fuel. That fuel energy does not all become useful power at the wheels. Instead, it splits into brake power (useful output), heat carried away by the exhaust, heat absorbed by the coolant, heat absorbed by the engine oil, and small losses to radiation and friction.

With a turbocharger, some exhaust energy is used to spin the turbine, which drives the compressor. That compressor increases the temperature of the intake air, and this heat must later be removed by the intercooler. As a result, a turbocharged cooling system must manage three primary heat loads: engine coolant heat (radiator), engine oil heat (oil cooler), compressed intake air heat (intercooler)

We will begin the analysis at the source of the energy, the E85 fuel. Fuel provides energy at a rate given by:

$$\dot{Q}_{fuel} = \dot{m}_f LHV$$

Taking an estimated flow rate of 6.55×10^{-3} kg/s at wide open throttle and 11,500 RPM with a Lower Heating Value (LHV) of about 29.4 MJ/kg for E85, we can estimate the total fuel energy entering the system per second.

$$\dot{Q}_{fuel} = \left(6.55 \times 10^{-3} \frac{kg}{s}\right) \left(29.4 \times 10^6 \frac{J}{kg}\right) \approx 193 \text{ kW}$$

This represents the total chemical energy delivered from fuel to the engine per second at peak load. Additionally, the engine is estimated to produce:

$$69 \text{ HP}_{max} = 51.5 \text{ kW}$$

This is the maximum mechanical power delivered at the crankshaft, representing the useful portion of the fuel energy. However, for this analysis, we will use the average horsepower from the provided endurance run data, which was 17 HP.

$$17 \text{ HP}_{avg} = 12.5 \text{ kW}$$

The exhaust mass flow rate can be approximated as the sum of the incoming air and fuel mass flow rates:

$$\dot{m}_{exh} \approx \dot{m}_{air} + \dot{m}_f$$

At maximum engine speed, the measured air mass flow rate is:

$$\dot{m}_{air} = 0.050823 \text{ kg/s}$$

Therefore,

$$\dot{m}_{exh} = 0.050823 + 0.00655$$

$$\dot{m}_{exh} = 0.05737 \text{ kg/s}$$

The sensible heat carried away by the exhaust gases can be estimated using:

$$\dot{Q}_{exhaust} = \dot{m}_{exh} c_p (T_{exh} - T_{ref})$$

For combustion products at high temperature, a representative value is:

$$c_p \approx 1.10 \frac{kJ}{kg \cdot K}$$

The maximum measured exhaust temperature downstream of the turbocharger is

$$T_{exh} = 1450^\circ F$$

which corresponds to

$$T_{exh} = 1061 K$$

Assuming an ambient reference temperature of

$$T_{ref} = 305 K$$

The temperature difference is

$$\Delta T = 756 K$$

Substituting:

$$\dot{Q}_{exhaust} = (0.05737)(1100)(756)$$

$$\dot{Q}_{exhaust} \approx 47.7 \text{ kW}$$

A Steady-state energy split for the turbocharged Yamaha R3 can be expressed as:

$$\dot{Q}_{fuel} = \dot{W}_b + \dot{Q}_{coolant} + \dot{Q}_{exhaust} + \dot{Q}_{oil} + \dot{Q}_{other}$$

Rearranging to isolate the coolant requirement:

$$\dot{Q}_{coolant} = \dot{Q}_{fuel} - \dot{W}_b - \dot{Q}_{exhaust} - (\dot{Q}_{oil} + \dot{Q}_{other})$$

For spark-ignition engines operating under boost, experimental studies and dynamometer data typically show that approximately 20-30% of the fuel's chemical energy is rejected through the coolant system. The remainder of the energy leaves through the exhaust stream, useful mechanical work, oil cooling, and other minor losses.

Using a representative value of 25% coolant heat rejection, the heat transferred to the coolant can be estimated as

$$\dot{Q}_{coolant} \approx 0.25 \dot{Q}_{fuel}$$

Substituting the previously calculated fuel energy rate,

$$\dot{Q}_{coolant} = 0.25(193)$$

$$\dot{Q}_{coolant} \approx 48.3 \text{ kW}$$

This value represents the approximate heat load that must be removed by the engine cooling system at peak operations.

Because the F26 racecar utilized two radiators operating in parallel, the heat rejection load is assumed to be evenly distributed between the two units. The required heat rejection capacity per radiator is therefore

$$\dot{Q}_{rad} = \frac{\dot{Q}_{coolant}}{2}$$

$$\dot{Q}_{rad} = \frac{48.3}{2}$$

$$\dot{Q}_{rad} = 24.1 \text{ kW}$$

Thus, each radiator must be capable of rejecting approximately 25 kW of thermal energy under maximum engine load conditions. This value establishes the thermal design requirement for the cooling system and serves as the basis for radiator sizing and airflow calculations used in the remainder of this analysis.

Appendix B – MoTeC Interface



Appendix C – Fin Designs

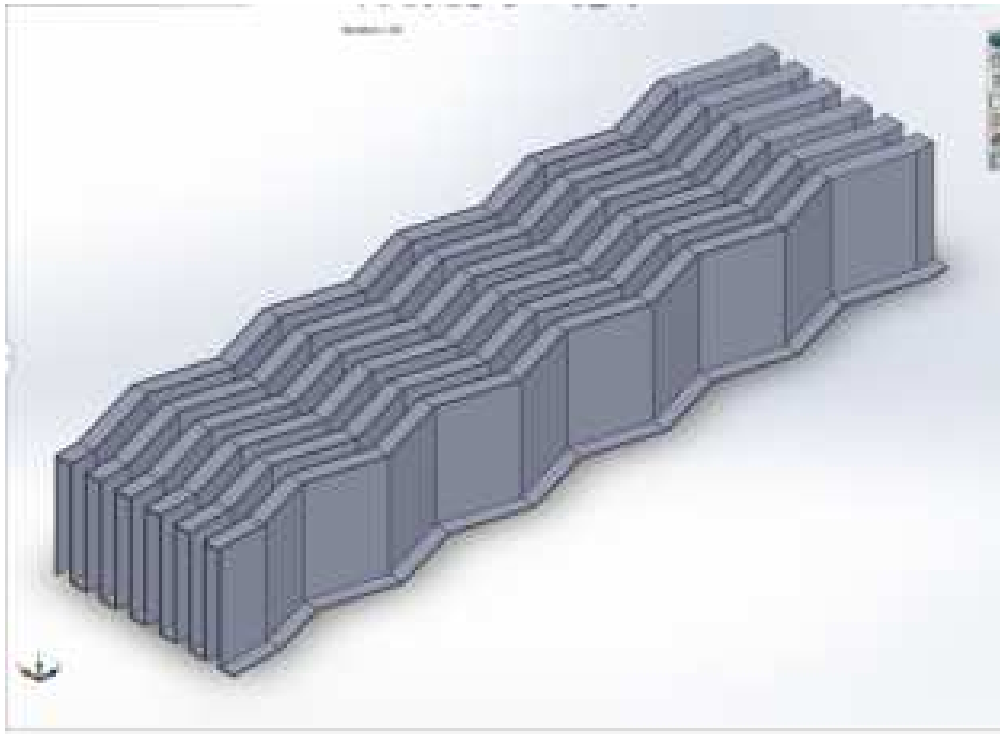


Figure 27. Wavy Fins Model

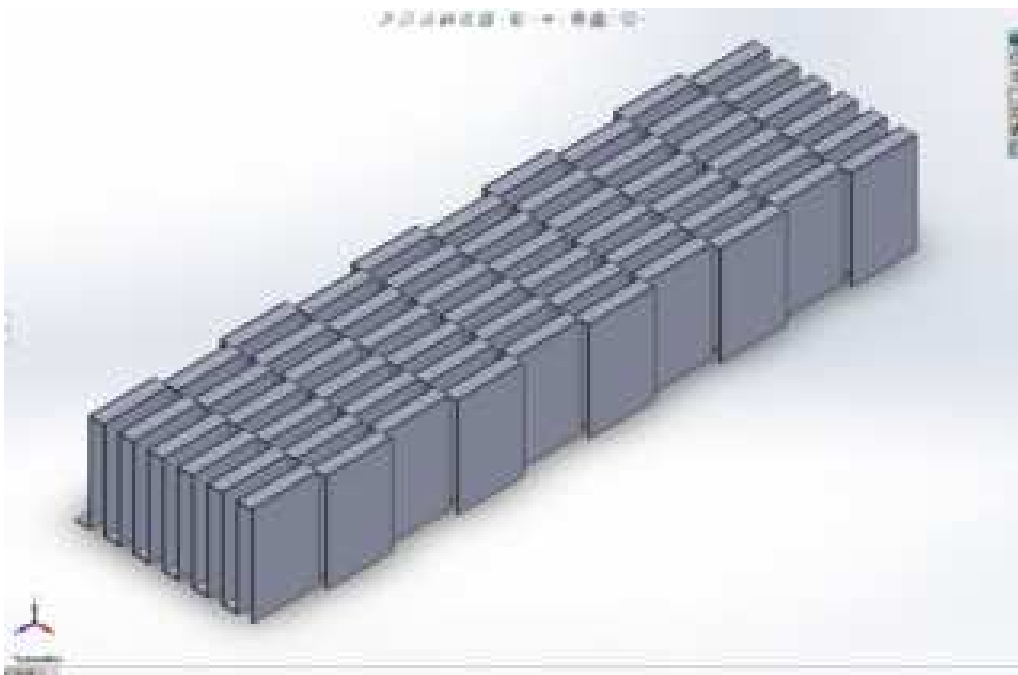


Figure 28. Offset Fins Model

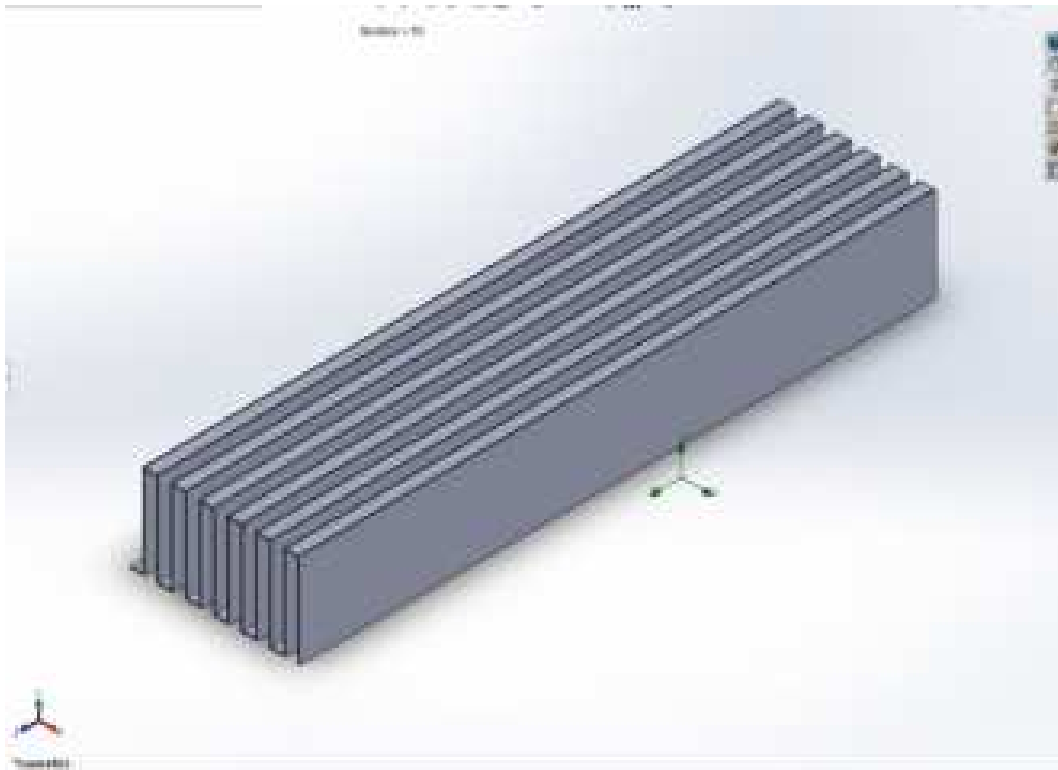


Figure 29. Straight Fins Model

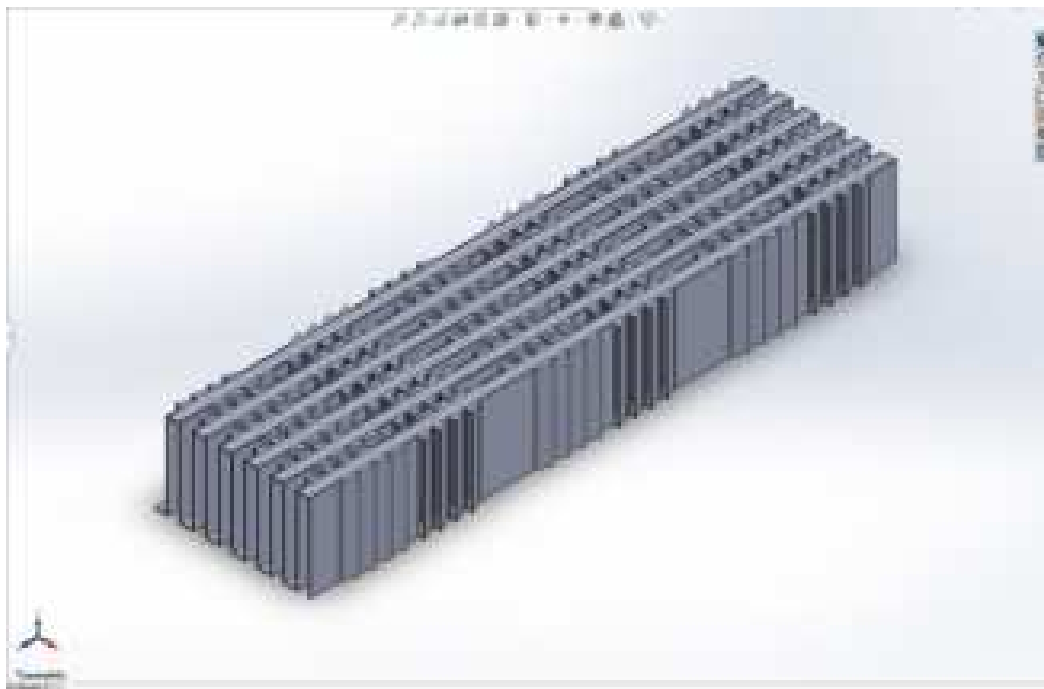


Figure 30. Louvered Fins Model

Appendix D – NTU-Effectiveness Method

To being the NTU method analysis, the geometries must be calculated for a full global analysis of the radiator. Data from Table I will be utilized for these equations from Section III. The tube pitch and air gap will vary slightly between the radiator geometries, since the height to number of tubes ratio is not the same.

$$p_{tube_{previous}} = \frac{H_{previous}}{N_{tubes}} = \frac{12 \text{ in}}{24} = 0.5 \text{ in}$$

$$p_{tube_{new}} = \frac{6 \text{ in}}{16} = 0.375 \text{ in}$$

The air gap between the fins can be found using the difference between the pitch of the tube and the height of the tube

$$H_{air} = p_{tube} - H_{tube}$$

On the air side, the hydraulic diameter can be calculated as follows, which will be important for calculating the Reynolds' number later. Note that W_{air} is the gap between fins

$$D_{h,air} = \frac{4H_{air}W_{air}}{2(H_{air} + W_{air})}$$

$$P_{fin} = \frac{0.0254}{FPI} = \frac{0.0254}{15} = 1.693 \text{ mm}$$

$$W_{air} = P_{fin} - t_{fin} = 1.693 - 0.1 = 1.593 \text{ mm}$$

From this, the fraction of the face area open to airflow can be calculated. Let the frontal area ratio be σ . This will be important later for computing the mass velocity.

$$\sigma = \frac{H_{air} W_{air}}{p_{tube} P_{fin}}$$

The total air-side heat transfer area accounts for the fin surface area and bare tube area between fins and is in terms of per unit face area.

$$A_{ht} = (2N_{fins}L_{fin} + N_{fins}(P_{fin} - t_{fin}))A_{face}$$

Note that the first term in the equation above is the two-sided fin area, and the second term is the bare tube wall exposed between the fins. In this case, L_{fin} is equal to H_{air} , which is the fin height. The inner tube wall area can be defined as below, where this is the actual surface where heat transfers from the coolant to the metal. This will be important for the water-side convection calculations. The tubes are assumed to be square for the simplification of the rounded edge geometry. W is the width of the radiator, in other words, the length of the tubes across the core.

$$P_{tube} = 2(D_{core} + H_{tube})$$

$$A_{tube,inner} = P_{tube}N_{tubes}W$$

The air-side convection coefficient can be calculated using the Colburn j-factor, which is outlined in Kays & London's wavy fin surface data. Tabulated values can be interpolated with the 15 FPI parameter [3].

$$j_{15} = (1 - w)j_{11.5} + wj_{17.8}$$

$$w = \frac{15 - 11.5}{17.8 - 11.5} = 0.56$$

We can also calculate the velocity through the contracted air passages, using the effective velocity. The effective air velocity represents the combined fan and ram air contribution through the core during a race. It is fixed at 2.0-2.5 m/s (4.5-5.5 mph), which is calculated using the minimum cfm outlined by Ramin's Cooling System and Design

Report. There will most likely be a little bit of air that gets into the side pod while driving, so a V_{eff} of 2.9 m/s (6.5 mph) will be used.

$$G = \frac{\rho_{air} V_{eff}}{\sigma}$$

The air-side Reynolds' number can then be calculated as such:

$$Re_{air} = \frac{G D_{h,air}}{\mu_{air}}$$

The air-side convective coefficient can be computed using the Colburn analogy, which integrates the j-factor. This j-factor is crucial for accounting for the surface geometry's effect on heat transfer on the air-side. Both radiators use the same fin geometry, so the same j-factor can be applied to both. G will differ slightly due to the different frontal area ratio (σ), stemming from the difference in tube pitch.

$$h_{air} = j(Re_{air}) G C_{p,air} Pr_{air}^{-\frac{2}{3}}$$

The fins play a large role in the heat transfer of the radiator. Thus, fin efficiency needs to be determined. Fins do not transfer heat uniformly from base to tip, and heat must conduct along the fin length, so the tip will run cooler than the base. This is very applicable at higher air velocities where the surface demand is high. The heat at the base will not have time to make it through the length of the tip before being cooled, decreasing the fin efficiency. For one fin, the efficiency equation is as follows.

$$\eta_{fin} = \frac{\tanh(mL)}{mL}$$

Where m is the fin parameter (units are in^{-1}). It represents how quickly the fin temperature decays from base to tip. A larger m means the tip runs much cooler than the base, resulting in an overall less effective fin.

$$m = \sqrt{\frac{2H_{air}}{k_{Al} t_{fin}}}$$

The overall surface efficiency accounts for the fraction of total air-side area that is fin versus only bare tube [7].

$$\eta_o = 1 - A_{fin,frac}(1 - \eta_{fin})$$

$A_{fin,frac}$ is approximately 0.86 and represents the fraction of total air-side area that is fin surface. This is determined from K&L data at 15 FPI. The effective air-side coefficient entering the calculation for the overall convective heat transfer coefficient will be as follows [3]:

$$h_{air,eff} = \eta_o h_{air}$$

It is important to note that as air velocity increases, h_{air} will also increase, and thus increase m. This will decrease the fin efficiency and may have diminishing returns. The heat capacity curve bends over at higher velocities not only because of NTU saturation but also because fins become less efficient as air-side demand increases. Keeping η_{fin} as close to 1 as possible is ideal. Next, the water-side heat transfer must be analyzed. The waterside Nusselt number will use the Dittus-Boelter correlation, also seen as equation 8.60 in Table 8.4 of Bergman, T.L., and Lavine, A.S.,

“*Fundamentals of Heat and Mass Transfer*” textbook. But first, the Reynolds’ number needs to be calculated. Note that v is the velocity of the coolant. P_{wet} is the wetted perimeter [7].

$$Re_w = \frac{\rho v D_{h,w}}{\mu_w}$$

$$D_{h,w} = \frac{4A_{x-sec}}{P_{wet}}$$

Reynolds’ number can be rewritten as a function of mass flow rate, substituting into the hydraulic diameter equation. The mass flow rate needs to be divided by the number of tubes per pass. A larger number of tubes will slow the mass flow rate through the core, allowing for the water to dissipate heat, as well as decreasing the heat capacity of the coolant.

$$Re_w = \frac{4 \left(\frac{\dot{m}}{N_{per\ pass}} \right)}{\mu_w P_{wet}}$$

The Dittus-Boelter correlation for the Nusselt number in turbulent flow is as follows. The flow is turbulent if the Reynolds’ number is greater than 4000 [5].

$$Nu_w = 0.023 Re_w^{0.8} Pr_w^{0.3}$$

If the flow is laminar ($Re_w < 4000$), then the Nusselt number can be defined as:

$$Nu_w = 3.66$$

However, due to the increased heat transfer rate that turbulent flow offers, our interest will only be geared towards the turbulent region. Turbulence in flow breaks down thermal boundary layers, offering much more effective heat transfer. From this, the water-side convective coefficient can now be calculated.

$$h_w = \frac{Nu_w k_w}{D_{h,tube}}$$

The properties of water (C_p , μ , k , Pr) vary significantly with temperature. Viscosity drops roughly 30% between 190 °F and 220 °F. Evaluating at the inlet temperature alone overstates resistance and underpredicts the convective coefficients. Properties are therefore to be evaluated at the bulk mean temperature, using the input coolant temperature and the target temperature at the outlet for a reasonable approximation.

$$T_{bulk} = \frac{T_{rad,in} + T_{target}}{2} = \frac{223^\circ\text{F} + 190^\circ\text{F}}{2} = 206.5^\circ\text{F}$$

Ultimately, airside will dominate the thermal resistance. The coolant outlet temperature is primarily governed by the mass flow rate through the equation below:

$$T_{rad,out} = T_{rad,in} - \frac{Q_{reject}}{\dot{m}C_p}$$

Thus, the design mass flow rate through the system must not pump water through the system too fast, since the coolant will not have enough time to dissipate the heat. The overall heat transfer coefficient, U , can be calculated,

which is essential for the NTU method. Because the airside and waterside areas are different, fins multiply the airside beyond the bare tube wall. Thus, the waterside resistance must be scaled by the area ratio AR.

$$\frac{1}{U} = \frac{1}{h_{air,eff}} + \frac{AR}{h_w}$$

Here are the values calculated at operating conditions:

$h_{air,eff}$	$\sim 150 \frac{W}{m^2 K}$
h_w	$\sim 2000 \frac{W}{m^2 K}$
AR	~ 12
Air-side resistance: $\frac{1}{h_{air,eff}}$	$0.00667 \frac{m^2 K}{W}$
Water-side resistance: $\frac{AR}{h_w}$	$0.00600 \frac{m^2 K}{W}$
Air-side share of total resistance	$\sim 53\%$
Water-side share of total resistance	$\sim 47\%$

Now, the NTU-effectiveness method can be applied to analyze the heat exchanger performance. The NTU method allows for the analysis of a heat exchanger without iterating the outlet temperatures. The heat capacity rates can be defined for the hot fluid (coolant) and the cold fluid (air).

$$C_{hot} = \dot{m}_w C_{p,w}$$

$$C_{cold} = \rho_{air} V_{eff} A_{face} C_{p,air}$$

$$C_{min} = \min(C_{hot}, C_{cold})$$

Where the heat capacity ratio is

$$C_r = \frac{C_{min}}{C_{max}}$$

NTU is a dimensionless measure of the heat exchanger's thermal size, in other words, how many times it can transfer the energy of the minimum fluid stream.

$$NTU = \frac{UA_{ht}}{C_{min}}$$

A higher NTU means the exchanger approaches thermodynamic maximum performances, and increases with larger area, higher U, or lower C_{min} . The effectiveness is the ratio of actual heat transfer to the thermodynamic maximum possible.

$$\varepsilon = \frac{Q_{actual}}{Q_{max}} = \frac{Q_{actual}}{C_{min}(T_{hot,in} - T_{cold,in})}$$

For a double-pass counter-crossflow arrangement, Shah & Sekulic define that the effectiveness per pass can be computed using $NTU_p = NTU/2$ and then can be combined. The effectiveness per pass follows this relation:

$$\varepsilon_p = 1 - \exp\left[\left(\frac{NTU_p^{0.22}}{C_r}\right) (\exp(-C_r NTU_p^{0.78}) - 1)\right]$$

Combining these for two passes:

$$F = \frac{1 - C_r P_1}{1 - P_1}$$

$$\varepsilon = \frac{F^2 - 1}{F^2 - C_r}$$

Therefore, the actual heat rejection and outlet temperatures will be as follows:

$$Q = \varepsilon C_{min}(T_{rad,in} - T_{atm})$$

$$T_{rad,out} = T_{rad,in} - \frac{Q}{C_{hot}}$$

$$T_{air,out} = T_{atm} + \frac{Q}{C_{cold}}$$

T_{air} will decrease as V_{eff} increases, though intuition might tell otherwise due to Q increasing. This is because C_{cold} will grow faster than Q . More air mass will pass through the core, and each unit of air picks up less heat [4].

Appendix E – NTU-Effectiveness Method MATLAB Simulation

Please note that the MATLAB script uses an Excel sheet to define the properties of both water and air. Please reach out to me at rmr6905@mavs.uta.edu for this Excel sheet if in need of running the script.

% FSAE Radiator Comparison - Previous (12x9 in) vs Current (Dual 9x6 in)
% Average condition only

clc; clear all

F2K = @(F) (F - 32)*5/9 + 273.15;
K2F = @(K) (K - 273.15)*9/5 + 32;
ms2mph = @(v) v * 2.237;

%% --- Inputs ---

% Engine

HP_avg = 20;
eta_heat = 0.30;
Q_avg = eta_heat * HP_avg * 745.7; % W

% Temperatures (endurance race measured)

T_atm_avg = F2K(80);
T_rad_in_avg = F2K(223);
T_air_out_meas = F2K(189); % measured on old radiator

% Fan

V_fan = 2.032; % m/s (150 ft/min bench test)

% Fan curve

Q_fan_m3s = [0, 50, 100, 150, 200, 250, 300, 350, 390] / 3600;
dP_fan_Pa = [188, 158, 133, 102, 80, 78, 61, 30, 0];

% V_eff switch

```

use_fixed_Veff = true;
V_eff_fixed    = 2.9;    % m/s

% Radiator geometry - shared construction
D_core = 1.25 * 0.0254;    % m flow depth
h_tube = 2e-3;            % m tube height
FPI     = 15;
t_fin   = 0.0001;         % m fin thickness
k_Al    = 167;            % W/mK
n_pass  = 2;

% Previous radiator (12 x 9 in)
W_old    = 12 * 0.0254;
H_old    = 9 * 0.0254;
N_tubes_old = 32;
N_per_pass_old = 16;

% New radiator (9 x 6 in)
W_new    = 9.0 * 0.0254;
H_new    = 6.0 * 0.0254;
N_tubes_new = 16;
N_per_pass_new = 8;

% RPM operating point
RPM_avg = 8104;

%% --- K&L Wavy Fin Surface Data (Kays & London 1984, p.257) ---
KL_115.fpi    = 11.5;
KL_115.Dh     = 3.023e-3;
KL_115.beta   = 1138;
KL_115.fin_frac = 0.822;
KL_115.Re = [200, 300, 400, 600, 800, 1000, 1500, 2000, 3000];
KL_115.j  = [0.060, 0.050, 0.043, 0.036, 0.031, 0.027, 0.022, 0.019, 0.016];
KL_115.f  = [0.220, 0.180, 0.160, 0.130, 0.115, 0.100, 0.085, 0.075, 0.062];

KL_178.fpi    = 17.8;
KL_178.Dh     = 2.123e-3;
KL_178.beta   = 1686;
KL_178.fin_frac = 0.892;
KL_178.Re = [200, 300, 400, 600, 800, 1000, 1500, 2000, 3000];
KL_178.j  = [0.070, 0.057, 0.049, 0.040, 0.034, 0.030, 0.025, 0.022, 0.018];
KL_178.f  = [0.250, 0.210, 0.180, 0.150, 0.130, 0.120, 0.100, 0.090, 0.075];

% Interpolate to 15 FPI
w = (FPI - KL_115.fpi) / (KL_178.fpi - KL_115.fpi);
KL_15.fpi    = FPI;
KL_15.Dh     = (1-w)*KL_115.Dh + w*KL_178.Dh;
KL_15.beta   = (1-w)*KL_115.beta + w*KL_178.beta;
KL_15.fin_frac = (1-w)*KL_115.fin_frac + w*KL_178.fin_frac;
KL_15.Re     = KL_115.Re;
KL_15.j      = (1-w)*KL_115.j + w*KL_178.j;
KL_15.f      = (1-w)*KL_115.f + w*KL_178.f;

j_fn = @(Re) interp1(KL_15.Re, KL_15.j, Re, 'pchip', 'extrap');
f_fn = @(Re) interp1(KL_15.Re, KL_15.f, Re, 'pchip', 'extrap');

```

```

%% --- Fluid Properties from Excel ---
opts_air = detectImportOptions('properties.xlsx', 'Sheet', 'Air');
opts_air.DataRange = 'B3:K37';
opts_air.VariableNames = {'T', 'rho', 'Cp', 'mew', 'v_raw', 'k_raw', ...
    'k_formula', 'alpha_raw', 'alpha_formula', 'Pr'};
air_tbl = readtable('properties.xlsx', opts_air);
T_av = air_tbl.T;
rho_air_fn = @(T) interp1(T_av, air_tbl.rho, T, 'pchip', 'extrap');
Cp_air_fn = @(T) interp1(T_av, air_tbl.Cp, T, 'pchip', 'extrap');
mu_air_fn = @(T) interp1(T_av, air_tbl.mew, T, 'pchip', 'extrap');
Pr_air_fn = @(T) interp1(T_av, air_tbl.Pr, T, 'pchip', 'extrap');

opts_wtr = detectImportOptions('properties.xlsx', 'Sheet', 'Water');
opts_wtr.DataRange = 'B5:Q58';
opts_wtr.VariableNames = {'T', 'P_bar', 'rho_liq', 'sv', 'h_fg', 'Cp_liq', ...
    'Cp_vap', 'mu_liq', 'mu_vap', 'k_liq', 'k_vap', ...
    'Pr_liq', 'Pr_vap', 'sigma_w', 'beta_w', 'T2'};
wtr_tbl = readtable('properties.xlsx', opts_wtr);
T_wv = wtr_tbl.T;
valid = ~isnan(T_wv) & ~isnan(wtr_tbl.Cp_liq) & ~isnan(wtr_tbl.mu_liq) & ...
    ~isnan(wtr_tbl.k_liq) & ~isnan(wtr_tbl.Pr_liq);
T_wv = T_wv(valid);
Cp_w_fn = @(T) interp1(T_wv, wtr_tbl.Cp_liq(valid), T, 'pchip', 'extrap');
mu_w_fn = @(T) interp1(T_wv, wtr_tbl.mu_liq(valid), T, 'pchip', 'extrap');
k_w_fn = @(T) interp1(T_wv, wtr_tbl.k_liq(valid), T, 'pchip', 'extrap');
Pr_w_fn = @(T) interp1(T_wv, wtr_tbl.Pr_liq(valid), T, 'pchip', 'extrap');

%% --- Pump Curve (mechanical pump, R3 test data) ---
RPM_pump_mech = [2000,3000,4000,5000,6000,7000,8000,9000,10000,10500,11000];
GPM_pump_mech = [1.35,2.00,2.85,3.50,4.30,5.10,5.95,6.50, 7.35, 7.40, 7.65];

gpm2kgs = @(gpm) gpm * 3.785e-3/60 * 1000;
mdot_mech_fn = @(rpm) gpm2kgs(interp1(RPM_pump_mech, GPM_pump_mech, ...
    min(rpm, RPM_pump_mech(end)), 'pchip'));

m_dot_avg = mdot_mech_fn(RPM_avg);

%% --- Air and Water Bulk Properties ---
T_air_bulk = T_atm_avg + 5;
rho_air_blk = rho_air_fn(T_air_bulk);
Cp_air_blk = Cp_air_fn(T_air_bulk);
mu_air_blk = mu_air_fn(T_air_bulk);
Pr_air_blk = Pr_air_fn(T_air_bulk);

Perim_tube = 2*(D_core + h_tube);
A_xsec_tube = D_core * h_tube;
D_h_tube = 4*A_xsec_tube / Perim_tube;

% Water properties evaluated at bulk mean temp (T_in + T_out) / 2.
% T_out assumed 190 F from measured data
T_cool_out_assumed = F2K(190);
T_bulk_w = (T_rad_in_avg + T_cool_out_assumed) / 2;
Cp_w_avg = Cp_w_fn(T_bulk_w);
mu_w_avg = mu_w_fn(T_bulk_w);

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```

k_w_avg = k_w_fn(T_bulk_w);
Pr_w_avg = Pr_w_fn(T_bulk_w);
C_hot_avg = m_dot_avg * Cp_w_avg;

%% --- Find Effective Air Velocity V_eff ---
old_geo = build_geo(W_old, H_old, D_core, N_tubes_old, N_per_pass_old, ...
    h_tube, FPI, t_fin, KL_15);

if use_fixed_Veff
    V_eff = V_eff_fixed;
    fprintf('V_eff = %.2f m/s (fixed, %.1f mph)\n\n', V_eff, V_eff*2.237);
else
    T_air_resid = @(V) T_air_out_fn(V, old_geo, j_fn, k_Al, ...
        C_hot_avg, rho_air_blk, Cp_air_blk, mu_air_blk, Pr_air_blk, ...
        mu_w_avg, k_w_avg, Pr_w_avg, D_h_tube, Perim_tube, m_dot_avg, ...
        T_rad_in_avg, T_atm_avg) - T_air_out_meas;

    opts_fz = optimset('TolX', 1e-6, 'Display', 'off');
    V_lo = KL_15.Re(1) * old_geo.sigma * mu_air_blk / (rho_air_blk * old_geo.Dh) *
1.05;
    V_hi = 40;
    res_lo = T_air_resid(V_lo);
    res_hi = T_air_resid(V_hi);

    if sign(res_lo) == sign(res_hi)
        fprintf('Back-calculation failed. Using V_eff_fixed = %.1f m/s\n\n',
V_eff_fixed);
        V_eff = V_eff_fixed;
    else
        V_eff = fzero(T_air_resid, [V_lo, V_hi], opts_fz);
        fprintf('V_eff back-calculated = %.2f m/s (%.1f mph)\n', V_eff, V_eff*2.237);
        fprintf('Fan = %.2f m/s, Ram = %.2f m/s\n\n', V_fan, V_eff - V_fan);
    end
end

%% --- Evaluate Both Radiators at V_eff ---
old = kl_rad_performance(W_old, H_old, D_core, N_tubes_old, N_per_pass_old, ...
    h_tube, FPI, t_fin, k_Al, V_eff, KL_15, j_fn, ...
    C_hot_avg, m_dot_avg, ...
    rho_air_blk, Cp_air_blk, mu_air_blk, Pr_air_blk, ...
    mu_w_avg, k_w_avg, Pr_w_avg, Cp_w_avg, ...
    D_h_tube, Perim_tube, T_rad_in_avg, T_atm_avg);

new = kl_rad_performance(W_new, H_new, D_core, N_tubes_new, N_per_pass_new, ...
    h_tube, FPI, t_fin, k_Al, V_eff, KL_15, j_fn, ...
    C_hot_avg, m_dot_avg, ...
    rho_air_blk, Cp_air_blk, mu_air_blk, Pr_air_blk, ...
    mu_w_avg, k_w_avg, Pr_w_avg, Cp_w_avg, ...
    D_h_tube, Perim_tube, T_rad_in_avg, T_atm_avg);

%% --- Dual New Radiators in Series (two 9x6 cores, fresh air each) ---
T_mid_avg = new.T_cool_out_avg;
% Core 2 inlet = T_mid_avg; assumed outlet still ~190 F
T_bulk_w2 = (T_mid_avg + T_cool_out_assumed) / 2;
Cp_w_mid_a = Cp_w_fn(T_bulk_w2);

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```

mu_w_mid_a = mu_w_fn(T_bulk_w2);
k_w_mid_a = k_w_fn(T_bulk_w2);
Pr_w_mid_a = Pr_w_fn(T_bulk_w2);
C_hot_mid_a = m_dot_avg * Cp_w_mid_a;

new2 = kl_rad_performance(W_new, H_new, D_core, N_tubes_new, N_per_pass_new, ...
    h_tube, FPI, t_fin, k_Al, V_eff, KL_15, j_fn, ...
    C_hot_mid_a, m_dot_avg, ...
    rho_air_blk, Cp_air_blk, mu_air_blk, Pr_air_blk, ...
    mu_w_mid_a, k_w_mid_a, Pr_w_mid_a, Cp_w_mid_a, ...
    D_h_tube, Perim_tube, T_mid_avg, T_atm_avg);

T_final_avg = new2.T_cool_out_avg;
Q_total_avg = C_hot_avg * (T_rad_in_avg - T_final_avg);

T_air_err_F = K2F(old.T_air_out_avg) - K2F(T_air_out_meas);

%% --- print results
fprintf('=====\\n');
fprintf(' Radiator Comparison (V_eff = %.1f mph) - Avg Condition\\n',
    ms2mph(V_eff));
fprintf(' Previous: 12x9 in | Current: Dual 2x(9x6) in series\\n');
fprintf('=====\\n\\n');

fprintf('Performance Summary (Avg: %.0f hp, %.0f W required)\\n', HP_avg, Q_avg);
fprintf(' %-32s %10s %10s %10s\\n', '', 'Previous', 'Current', 'Dual Current');
fprintf(' %-32s %8.0f W %8.0f W %8.0f W\\n', 'Heat capacity', ...
    old.Q_cap_avg, new.Q_cap_avg, Q_total_avg);
fprintf(' %-32s %8.1f%% %8.1f%% %8.1f%%\\n', 'Margin over required', ...
    (old.Q_cap_avg-Q_avg)/Q_avg*100, (new.Q_cap_avg-Q_avg)/Q_avg*100, ...
    (Q_total_avg-Q_avg)/Q_avg*100);
fprintf(' %-32s %8.1f F %8.1f F %8.1f F\\n', 'Coolant outlet', ...
    K2F(old.T_cool_out_avg), K2F(new.T_cool_out_avg), K2F(T_final_avg));
fprintf(' %-32s %8.1f F %8.1f F\\n', 'Air outlet', ...
    K2F(old.T_air_out_avg), K2F(new.T_air_out_avg));

%% =====
% Sweep data - air velocity
% =====
V_sweep = linspace(0.5, 25, 150);
Q_old_V = zeros(size(V_sweep));
Q_new_V = zeros(size(V_sweep));
T_cout_dual_V = zeros(size(V_sweep));

for i = 1:length(V_sweep)
    Vi = V_sweep(i);

    Q_old_V(i) = Q_at_V(Vi, old, j_fn, k_Al, C_hot_avg, ...
        rho_air_blk, Cp_air_blk, mu_air_blk, Pr_air_blk, T_rad_in_avg, T_atm_avg);
    Q_new_V(i) = Q_at_V(Vi, new, j_fn, k_Al, C_hot_avg, ...
        rho_air_blk, Cp_air_blk, mu_air_blk, Pr_air_blk, T_rad_in_avg, T_atm_avg);

    % Dual new in series at this velocity
    G_i = rho_air_blk * Vi / new.sigma;
    Re_i = G_i * new.Dh / mu_air_blk;

```



```

j_i    = j_fn(Re_i);
h_ai   = j_i * G_i * Cp_air_blk * Pr_air_blk^(-2/3);
mL_i   = sqrt(2*h_ai/(k_A1*t_fin)) * new.L_fin;
ef_i   = tanh(mL_i)/mL_i;
eo_i   = 1 - new.A_fin_frac*(1-ef_i);
heff_i = eo_i * h_ai;
C_ci   = rho_air_blk * Vi * new.A_face * Cp_air_blk;

% Core 1
Re_w1  = 4*(m_dot_avg/N_per_pass_new)/(Perim_tube*mu_w_avg);
Nu_w1  = 3.66; if Re_w1>4000; Nu_w1=0.023*Re_w1^0.8*Pr_w_avg^0.3; end
U1     = 1/(1/heff_i + new.AR/(Nu_w1*k_w_avg/D_h_tube));
C_mn   = min(C_hot_avg, C_ci);
eps1   = dbl_pass_eps(U1*new.A_ht/C_mn, C_mn/max(C_hot_avg,C_ci));
Q1     = eps1 * C_mn * (T_rad_in_avg - T_atm_avg);
T_m    = T_rad_in_avg - Q1/C_hot_avg;

% Core 2 - bulk mean: core-2 inlet is T_m, assumed outlet ~190 F
T_blk2 = (T_m + T_cool_out_assumed) / 2;
Cp_m   = Cp_w_fn(T_blk2); mu_m = mu_w_fn(T_blk2);
k_m    = k_w_fn(T_blk2); Pr_m = Pr_w_fn(T_blk2);
C_hm   = m_dot_avg * Cp_m;
Re_w2  = 4*(m_dot_avg/N_per_pass_new)/(Perim_tube*mu_m);
Nu_w2  = 3.66; if Re_w2>4000; Nu_w2=0.023*Re_w2^0.8*Pr_m^0.3; end
U2     = 1/(1/heff_i + new.AR/(Nu_w2*k_m/D_h_tube));
C_mn2  = min(C_hm, C_ci);
eps2   = dbl_pass_eps(U2*new.A_ht/C_mn2, C_mn2/max(C_hm,C_ci));
Q2     = eps2 * C_mn2 * (T_m - T_atm_avg);
T_cout_dual_V(i) = K2F(T_m - Q2/C_hm);
end

T_cout_old_V = K2F(T_rad_in_avg - Q_old_V/C_hot_avg);
T_air_old_V  = K2F(T_atm_avg + Q_old_V ./
(rho_air_blk*V_sweep*old.A_face*Cp_air_blk));
T_air_new_V  = K2F(T_atm_avg + Q_new_V ./
(rho_air_blk*V_sweep*new.A_face*Cp_air_blk));

%% =====
% Sweep data - mass flow rate
% =====
mdot_sweep = linspace(0.05, 0.55, 150);
T_cout_old_M = zeros(size(mdot_sweep));
T_cout_dual_M = zeros(size(mdot_sweep));
Q_old_M      = zeros(size(mdot_sweep));
Q_dual_M     = zeros(size(mdot_sweep));

C_ci_old = rho_air_blk * V_eff * old.A_face * Cp_air_blk;
C_ci_new = rho_air_blk * V_eff * new.A_face * Cp_air_blk;
heff_fix = new.eta_o * new.h_air; % air-side fixed at V_eff

for i = 1:length(mdot_sweep)
    md    = mdot_sweep(i);
    C_ha  = md * Cp_w_avg;

    % --- Old radiator ---

```

```

Re_wa = 4*(md/N_per_pass_old)/(Perim_tube*mu_w_avg);
Nu_wa = 3.66; if Re_wa>4000; Nu_wa=0.023*Re_wa^0.8*Pr_w_avg^0.3; end
U_oa = 1/(1/(old.eta_o*old.h_air) + old.AR/(Nu_wa*k_w_avg/D_h_tube));
C_mn = min(C_ha, C_ci_old);
eps = dbl_pass_eps(U_oa*old.A_ht/C_mn, C_mn/max(C_ha,C_ci_old));
Q_old_M(i) = eps * C_mn * (T_rad_in_avg - T_atm_avg);
T_cout_old_M(i) = K2F(T_rad_in_avg - Q_old_M(i)/C_ha);

% --- Dual new: core 1 ---
Re_w1 = 4*(md/N_per_pass_new)/(Perim_tube*mu_w_avg);
Nu_w1 = 3.66; if Re_w1>4000; Nu_w1=0.023*Re_w1^0.8*Pr_w_avg^0.3; end
U1 = 1/(1/heff_fix + new.AR/(Nu_w1*k_w_avg/D_h_tube));
C_mn = min(C_ha, C_ci_new);
eps1 = dbl_pass_eps(U1*new.A_ht/C_mn, C_mn/max(C_ha,C_ci_new));
Q1 = eps1 * C_mn * (T_rad_in_avg - T_atm_avg);
T_m = T_rad_in_avg - Q1/C_ha;

% Core 2 - bulk mean: core-2 inlet is T_m, assumed outlet ~190 F
T_blk2 = (T_m + T_cool_out_assumed) / 2;
Cp_m = Cp_w_fn(T_blk2); mu_m = mu_w_fn(T_blk2);
k_m = k_w_fn(T_blk2); Pr_m = Pr_w_fn(T_blk2);
C_hm = md * Cp_m;
Re_w2 = 4*(md/N_per_pass_new)/(Perim_tube*mu_m);
Nu_w2 = 3.66; if Re_w2>4000; Nu_w2=0.023*Re_w2^0.8*Pr_m^0.3; end
U2 = 1/(1/heff_fix + new.AR/(Nu_w2*k_m/D_h_tube));
C_mn2 = min(C_hm, C_ci_new);
eps2 = dbl_pass_eps(U2*new.A_ht/C_mn2, C_mn2/max(C_hm,C_ci_new));
Q2 = eps2 * C_mn2 * (T_m - T_atm_avg);
Q_dual_M(i) = Q1 + Q2;
T_cout_dual_M(i) = K2F(T_m - Q2/C_hm);
end

%% =====
% Plot 1: Heat Capacity vs Air Velocity
% =====
figure('Position',[50 50 900 500]);
plot(ms2mph(V_sweep), Q_old_V/1000, 'b-', 'LineWidth',2); hold on;
plot(ms2mph(V_sweep), Q_new_V/1000, 'r-', 'LineWidth',2);
yline(Q_avg/1000, 'k:', sprintf('Req avg %.1f kW', Q_avg/1000), 'FontSize',8);
%xline(ms2mph(V_fan), 'g-', sprintf('V_{fan} %.1f mph', ms2mph(V_fan)),
'FontSize',8, 'LineWidth',1.5);
xline(ms2mph(V_eff), 'k--', sprintf('V_{eff} %.1f mph', ms2mph(V_eff)), 'FontSize',8,
'LineWidth',1.5);
xlabel('Air Velocity (mph)'); ylabel('Heat Capacity (kW)');
title(sprintf('Heat Capacity vs Air Velocity (V_{eff} = %.1f mph)', ms2mph(V_eff)));
legend('Previous (12x9)', 'Current (9x6)', 'Location', 'southeast');
grid on; box on;

%% =====
% Plot 2: Air Outlet Temperature vs Air Velocity
% =====
figure('Position',[100 50 900 480]);
plot(ms2mph(V_sweep), T_air_old_V, 'b-', 'LineWidth',2); hold on;
plot(ms2mph(V_sweep), T_air_new_V, 'r-', 'LineWidth',2);

```

```

yline(K2F(T_air_out_meas), 'b--', sprintf('Measured old %.0f F',
K2F(T_air_out_meas)), 'FontSize',9);
% xline(ms2mph(V_fan), 'g-', sprintf('V_{fan} %.1f mph', ms2mph(V_fan)),
'FontSize',8, 'LineWidth',1.5);
xline(ms2mph(V_eff), 'k--', sprintf('V_{eff} %.1f mph', ms2mph(V_eff)), 'FontSize',8,
'LineWidth',1.5);
xlabel('Air Velocity (mph)'); ylabel('Air Outlet Temp (F)');
title('Air Outlet Temperature vs Air Velocity (avg condition)');
legend('Previous (12x9)', 'Current (9x6)', 'Measured old', 'Location', 'northeast');
grid on; box on;

%% =====
% Plot 3: Coolant Outlet Temp vs Air Velocity
% =====
figure('Position',[150 100 950 500]);
plot(ms2mph(V_sweep), T_cout_old_V, 'b-', 'LineWidth',2); hold on;
plot(ms2mph(V_sweep), T_cout_dual_V, 'r-', 'LineWidth',2);
% xline(ms2mph(V_fan), 'g-', sprintf('V_{fan} %.1f mph', ms2mph(V_fan)),
'FontSize',8, 'LineWidth',1.5);
xline(ms2mph(V_eff), 'k--', sprintf('V_{eff} %.1f mph', ms2mph(V_eff)), 'FontSize',8,
'LineWidth',1.5);
xlabel('Air Velocity (mph)'); ylabel('Coolant Outlet Temp (deg F)');
title('Coolant Outlet Temp vs Air Velocity');
legend('Previous (12x9)', 'Current Dual 2x(9x6)', 'Location', 'northeast');
grid on; box on;
ylim([0 235]);

%% =====
% Plot 4: Previous Radiator vs Dual In Series - Coolant Outlet Temp vs Mass Flow
Rate
% =====
figure('Position',[200 150 950 500]);
plot(mdot_sweep, T_cout_old_M, 'b-', 'LineWidth',2); hold on;
plot(mdot_sweep, T_cout_dual_M, 'r-', 'LineWidth',2);
xline(m_dot_avg, 'k:', sprintf('Pump avg %.3f kg/s', m_dot_avg), 'FontSize',8);
xlabel('Coolant Mass Flow Rate (kg/s)');
ylabel('Coolant Outlet Temp (deg F)');
title(sprintf('Coolant Outlet Temp vs Mass Flow Rate (V_{eff} = %.1f mph)',
ms2mph(V_eff)));
legend('Previous Radiator (12x9)', 'Current Dual 2x(9x6)', 'Location', 'northeast');
grid on; box on;
xlim([0 0.7]);
ylim([0 235]);

%% =====
% Plot 5: Heat Capacity vs Coolant Mass Flow Rate
% =====
figure('Position',[250 200 950 500]);
plot(mdot_sweep, Q_old_M/1000, 'b-', 'LineWidth',2); hold on;
plot(mdot_sweep, Q_dual_M/1000, 'r-', 'LineWidth',2);
yline(Q_avg/1000, 'k:', sprintf('Required %.1f kW', Q_avg/1000), 'FontSize',8);
xline(m_dot_avg, 'k:', sprintf('Pump avg %.3f kg/s', m_dot_avg), 'FontSize',8);
xlabel('Coolant Mass Flow Rate (kg/s)');
ylabel('Heat Capacity (kW)');

```

```

title(sprintf('Heat Capacity vs Coolant Mass Flow Rate (V_{eff} = %.1f mph)',
ms2mph(V_eff)));
legend('Previous (12x9)', 'Current Dual (9x6)', 'Location', 'southeast');
grid on; box on;

%% =====
% Functions
%% =====

function g = build_geo(W, H, D_core, N_tubes, N_per_pass, h_tube, FPI, t_fin, KL_15)
    g.A_face      = W * H;
    g.tube_pitch  = H / N_tubes;
    g.h_air_pass  = g.tube_pitch - h_tube;
    g.L_fin       = g.h_air_pass;
    N_fins        = FPI / 0.0254;
    fin_pitch     = 1 / N_fins;
    w_air_pass    = fin_pitch - t_fin;
    g.Dh          = 4*(g.h_air_pass*w_air_pass) / (2*(g.h_air_pass+w_air_pass));
    g.sigma       = (g.h_air_pass/g.tube_pitch) * (w_air_pass/fin_pitch);
    A_fin_u_a     = 2*N_fins*g.L_fin;
    A_bare_u_a    = N_fins*(fin_pitch-t_fin);
    g.A_total_u_a = A_fin_u_a + A_bare_u_a;
    g.A_ht        = g.A_total_u_a * g.A_face;
    g.A_fin_frac  = KL_15.fin_frac;
    g.N_per_pass  = N_per_pass;
    % Inner tube wall area: Perim_inner * tube_length * N_tubes
    % Tubes run in the W direction; N_tubes total (all passes)
    g.A_tube_inner = 2*(D_core + h_tube) * N_tubes * W;
    g.AR          = g.A_ht / g.A_tube_inner; % air-to-tube area ratio
end

function T_out = T_air_out_fn(V, geo, j_fn, k_Al, C_hot, rho_air, ...
    Cp_air, mu_air, Pr_air, mu_w, k_w, Pr_w, D_h_tube, Perim_tube, m_dot_w, ...
    T_rad_in, T_atm)
    G      = rho_air * V / geo.sigma;
    Re_G   = G * geo.Dh / mu_air;
    j      = j_fn(Re_G);
    h_a    = j * G * Cp_air * Pr_air^(-2/3);
    mL     = sqrt(2*h_a/(k_Al*0.0001)) * geo.L_fin;
    ef     = tanh(mL)/mL;
    eo     = 1 - geo.A_fin_frac*(1-ef);
    Re_w   = 4*(m_dot_w/geo.N_per_pass) / (Perim_tube * mu_w);
    Nu_w   = 3.66; if Re_w > 4000; Nu_w = 0.023*Re_w^0.8*Pr_w^0.3; end
    h_w    = Nu_w * k_w / D_h_tube;
    U      = 1/(1/(eo*h_a) + geo.AR/h_w);
    C_cold = rho_air * V * geo.A_face * Cp_air;
    C_min  = min(C_hot, C_cold);
    Cr     = C_min / max(C_hot, C_cold);
    NTU    = U * geo.A_ht / C_min;
    eps    = dbl_pass_eps(NTU, Cr);
    Q      = eps * C_min * (T_rad_in - T_atm);
    T_out  = T_atm + Q / C_cold;
end

```

```

function Qa = Q_at_V(V, r, j_fn, k_Al, C_hot, rho_air, Cp_air, mu_air, Pr_air, ...
    T_rad_in, T_atm)
    G = rho_air * V / r.sigma;
    Re_G = G * r.Dh / mu_air;
    j = j_fn(Re_G);
    h_a = j * G * Cp_air * Pr_air^(-2/3);
    mL = sqrt(2*h_a/(k_Al*0.0001)) * r.L_fin;
    ef = tanh(mL)/mL;
    eo = 1 - r.A_fin_frac*(1-ef);
    U_a = 1/(1/(eo*h_a) + r.AR/r.h_water_avg);
    C_c = rho_air * V * r.A_face * Cp_air;
    C_mn = min(C_hot, C_c);
    eps = dbl_pass_eps(U_a*r.A_ht/C_mn, C_mn/max(C_hot,C_c));
    Qa = eps * C_mn * (T_rad_in - T_atm);
end

function r = kl_rad_performance(W, H, D_core, N_tubes, N_per_pass, ...
    h_tube, FPI, t_fin, k_Al, V, KL_15, j_fn, ...
    C_hot_avg, m_dot_avg, ...
    rho_air, Cp_air, mu_air, Pr_air, ...
    mu_w_avg, k_w_avg, Pr_w_avg, Cp_w_avg, ...
    D_h_tube, Perim_tube, T_rad_in_avg, T_atm_avg)

    r.A_face = W * H;
    r.tube_pitch = H / N_tubes;
    r.h_air_pass = r.tube_pitch - h_tube;
    r.L_fin = r.h_air_pass;
    N_fins = FPI / 0.0254;
    fin_pitch = 1 / N_fins;
    w_air_pass = fin_pitch - t_fin;
    r.Dh = 4*(r.h_air_pass*w_air_pass)/(2*(r.h_air_pass+w_air_pass));
    r.sigma = (r.h_air_pass/r.tube_pitch)*(w_air_pass/fin_pitch);
    A_fin_ua = 2*N_fins*r.L_fin;
    A_bare_ua = N_fins*(fin_pitch-t_fin);
    r.A_total_ua = A_fin_ua + A_bare_ua;
    r.A_ht = r.A_total_ua * r.A_face;
    r.A_fin_frac = KL_15.fin_frac;
    % Inner tube wall area and area ratio referred to air-side
    r.A_tube_inner = Perim_tube * N_tubes * W;
    r.AR = r.A_ht / r.A_tube_inner;

    r.G = rho_air * V / r.sigma;
    r.Re_G = r.G * r.Dh / mu_air;
    r.j_air = j_fn(r.Re_G);
    r.h_air = r.j_air * r.G * Cp_air * Pr_air^(-2/3);

    m_fin = sqrt(2*r.h_air/(k_Al*t_fin));
    mL = m_fin * r.L_fin;
    r.eta_f = tanh(mL)/mL;
    r.eta_o = 1 - r.A_fin_frac*(1-r.eta_f);
    h_eff = r.eta_o * r.h_air;

    Re_w_a = 4*(m_dot_avg/N_per_pass)/(Perim_tube*mu_w_avg);
    Nu_w_a = 3.66; if Re_w_a>4000; Nu_w_a=0.023*Re_w_a^0.8*Pr_w_avg^0.3; end
    r.h_water_avg = Nu_w_a * k_w_avg / D_h_tube;

```

```

r.U_avg = 1/(1/h_eff + r.AR/r.h_water_avg);
r.C_cold = rho_air * V * r.A_face * Cp_air;

C_min_a = min(C_hot_avg, r.C_cold);
r.NTU_avg = r.U_avg * r.A_ht / C_min_a;
r.eps_avg = dbl_pass_eps(r.NTU_avg, C_min_a/max(C_hot_avg,r.C_cold));
r.Q_cap_avg = r.eps_avg * C_min_a * (T_rad_in_avg - T_atm_avg);
r.T_cool_out_avg = T_rad_in_avg - r.Q_cap_avg / C_hot_avg;
r.T_air_out_avg = T_atm_avg + r.Q_cap_avg / r.C_cold;
end

function eps = dbl_pass_eps(NTU, Cr)
% Double-pass counter-crossflow effectiveness
% Shah & Sekulic eq. 3.131, N=2
eps_cf = @(n,c) 1 - exp((n.^0.22./c).*(exp(-c.*n.^0.78)-1));
NTU_p = NTU/2;
Cr_s = max(Cr, 1e-4);
P1 = eps_cf(NTU_p, Cr_s);
if Cr < 1e-4; P1 = 1 - exp(-NTU_p); end
if abs(1-Cr) < 1e-4
    eps = 2*P1/(1+P1);
else
    F = (1-Cr*P1)/(1-P1);
    eps = (F^2-1)/(F^2-Cr);
end
end

```

Appendix F – Geometric Output Equations

$$A_{t,cross} = (D - 2t_w)(H_{t,o} - 2t_w) + \frac{\pi}{4}(H_{t,o} - 2t_w)^2$$

$$P_o = 2(D) + \pi H_{t,o}$$

$$P_i = 2(D - 2t_w) + \pi(H_{t,o} - 2t_w)$$

$$A_{s,i} = T_{tubes} * P_i * W$$

$$A_{flow} = A_{t,cross} * Tubes/Pass$$

$$D_{h,h} = \frac{4 * A}{P}$$

$$A_{s,o} = T_{tubes} * P_o * W$$

$$A_{front} = W * H$$

$$A_{ff} = A_{front} * \phi_{ff}$$

$$p_{fins} = \frac{1}{Fins/mm}$$

$$T_{fins} = Fins/mm * W * (T_{tubes} - 1)$$

$$A_f = 2 * T_{fins} * D * L_f$$

$$A_{total} = A_{s.o} + A_f$$

$$D_{h,c} = 2 * (p_{fins} - t_f)$$

Appendix G – Fluid Output Equations

$$V_c = \frac{\dot{m}_c}{\rho_c A_{ff}}$$

$$V_h = \frac{\dot{m}_h}{\rho_h A_{flow}}$$

$$Re = \frac{\rho V D_h}{\mu}$$

$$Nu_c = 0.23 Re_c^{0.8} Pr_c^{0.4}$$

$$Nu_h = 0.23 Re_h^{0.8} Pr_h^{0.3}$$

$$h = \frac{Nu \cdot k}{D_h}$$

Appendix H – Pressure Output Equations

$$f_D = 0.3164 Re^{-0.25}$$

$$\Delta P_{tot,c} = f_{D,c} \left(\frac{D}{D_{h,c}} \right) \left(\frac{\rho_c V_c^2}{2} \right)$$

$$L_{eq} = N_{pass} W$$

$$\Delta P_{major,h} = f_{D,h} \left(\frac{L_{eq}}{D_{h,h}} \right) \left(\frac{\rho_h V_h^2}{2} \right)$$

$$\Delta P_{minor,h} = K \left(\frac{\rho_h V_h^2}{2} \right)$$

$$\Delta P_{tot,h} = \Delta P_{major,h} + \Delta P_{minor,h}$$

Appendix I – Extra Tool Sections

The Unit Conversion section, seen in Fig. 31, allows the engineer to convert all of the common parameters used in the tool between English and SI units. This allows the engineer to focus on sizing the radiator rather than having to exit the tool to find the values they need. The Information section, seen in Fig. 32, is intended not only to aid the engineer when certain checks fail, but also to grow as the engineer becomes more familiar with the tool

Parameter	English		SI	
Temperature	95	°F	35	°C
Length	8	in	203.20	mm
Power	180	HP	134.23	kW
	457227	BTU/hr	134.00	W
Volume Flow	180	CFM	0.085	m ³ /s
Density	1	lbm/ft ³	16.02	kg/m ³
Viscosity	1	lbf/ft·s	1.49	Pa·s
Pressure	1	psi	6894.76	Pa
	1	inH ₂ O	249	Pa
Fan Speed	180	CFM	0.10	kg/s

Parameter	SI		English	
Temperature	35	°C	95	°F
Length	203.2	mm	8	in
Power	134	kW	179.70	HP
			457227	BTU/hr
Volume Flow	0.085	m ³ /s	180.10	CFM
Density	5	kg/m ³	0.31	lbm/ft ³
Viscosity	3	Pa·s	2.02	lbf/ft·s
Pressure	6894.76	Pa	1	psi
			27.69	inH ₂ O
Fan Speed	0.1	kg/s	179.57	CFM

Figure 31. Unit Conversion Section

Q_{max} and Effectiveness Not Ok:

The maximum heat rejection of the system is lower than the required heat rejection

1. Increase $C_{\min}$
2. Lower required heat rejection rate
3. Increase air mass flow rate
4. Increase the inlet temperature difference

Figure 32. Information Section

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